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Takashima

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(54) **INFORMATION PROCESSING APPARATUS AND METHOD, INFORMATION RECORDING MEDIUM MANUFACTURING APPARATUS AND METHOD, AND INFORMATION RECORDING MEDIUM**

(75) Inventor: **Yoshikazu Takashima**, Tokyo (JP)

(73) Assignee: **Sony Corporation**, Tokyo (JP)

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(30) **Foreign Application Priority Data**

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(51) **Int. Cl.**
G06F 21/00 (2006.01)

(52) **U.S. Cl.**
USPC **726/27**

(58) **Field of Classification Search** 726/26
See application file for complete search history.

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Primary Examiner — Gilberto Barron, Jr.
Assistant Examiner — David Le
(74) *Attorney, Agent, or Firm* — Wolf, Greenfield & Sacks, P.C.

(57) **ABSTRACT**

An information processing apparatus includes a data processor that obtains a security check code included in content codes recorded on an information recording medium when using content and that conducts security check for the information processing apparatus in accordance with the obtained security check code. The data processor obtains information based on at least one of identification information and configuration information corresponding to the information processing apparatus or a content usage application, and among security check codes classified into a plurality of data files, the data processor selects a security check code corresponding to the obtained information to conduct security check on the basis of the selected security check code.

22 Claims, 16 Drawing Sheets

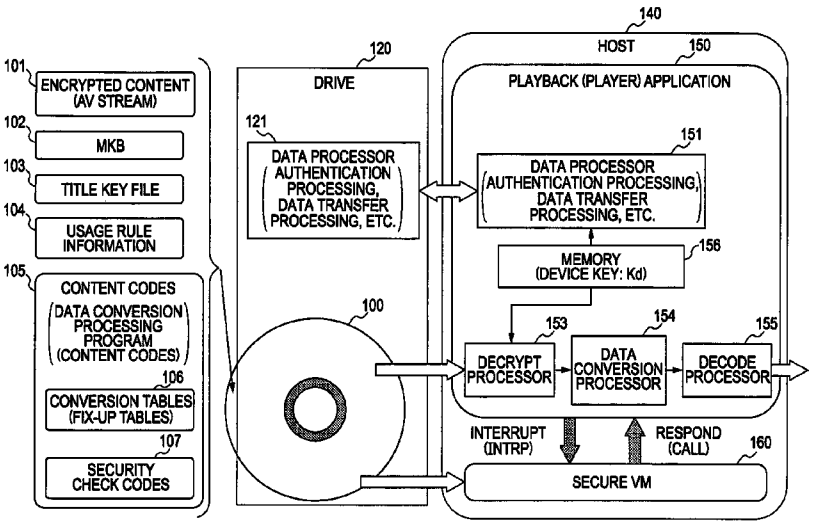


FIG. 1

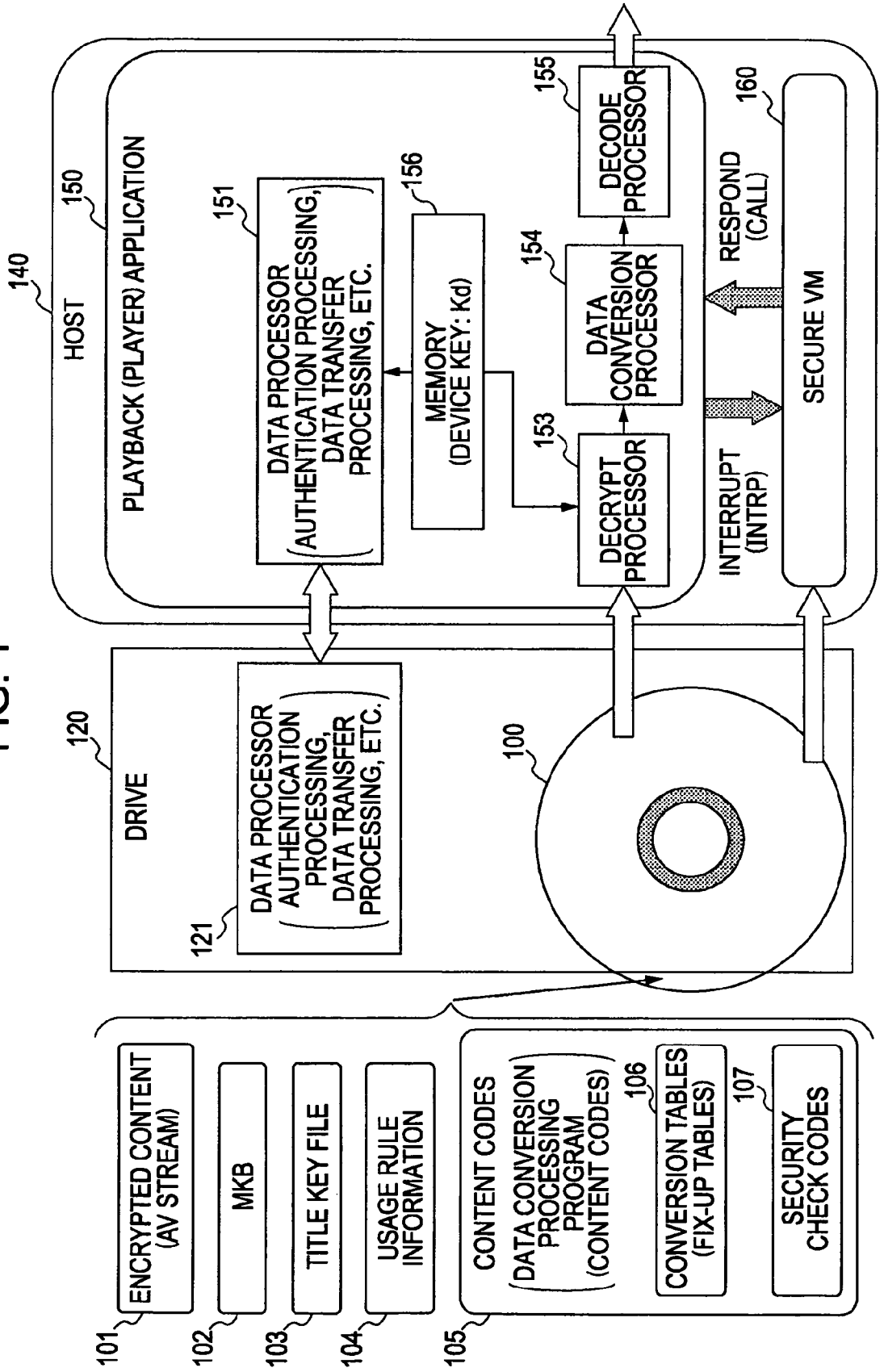

140
HOST
150
PLAYBACK (PLAYER) APPLICATION
151
DATA PROCESSOR (AUTHENTICATION PROCESSING, DATA TRANSFER PROCESSING, ETC.
156
MEMORY (DEVICE KEY: Kd)
153
DECRYPT PROCESSOR
154
DATA CONVERSION PROCESSOR
155
DECODE PROCESSOR
160
SECURE VM
RESPOND (CALL)
INTERRUPT (INTRP)
120
DRIVE
121
DATA PROCESSOR (AUTHENTICATION PROCESSING, DATA TRANSFER PROCESSING, ETC.)
100
101
ENCRYPTED CONTENT (AV STREAM)
102
MKB
103
TITLE KEY FILE
104
USAGE RULE INFORMATION
105
CONTENT CODES
DATA CONVERSION PROCESSING PROGRAM (CONTENT CODES)
106
CONVERSION TABLES (FIX-UP TABLES)
107
SECURITY CHECK CODES

FIG. 2

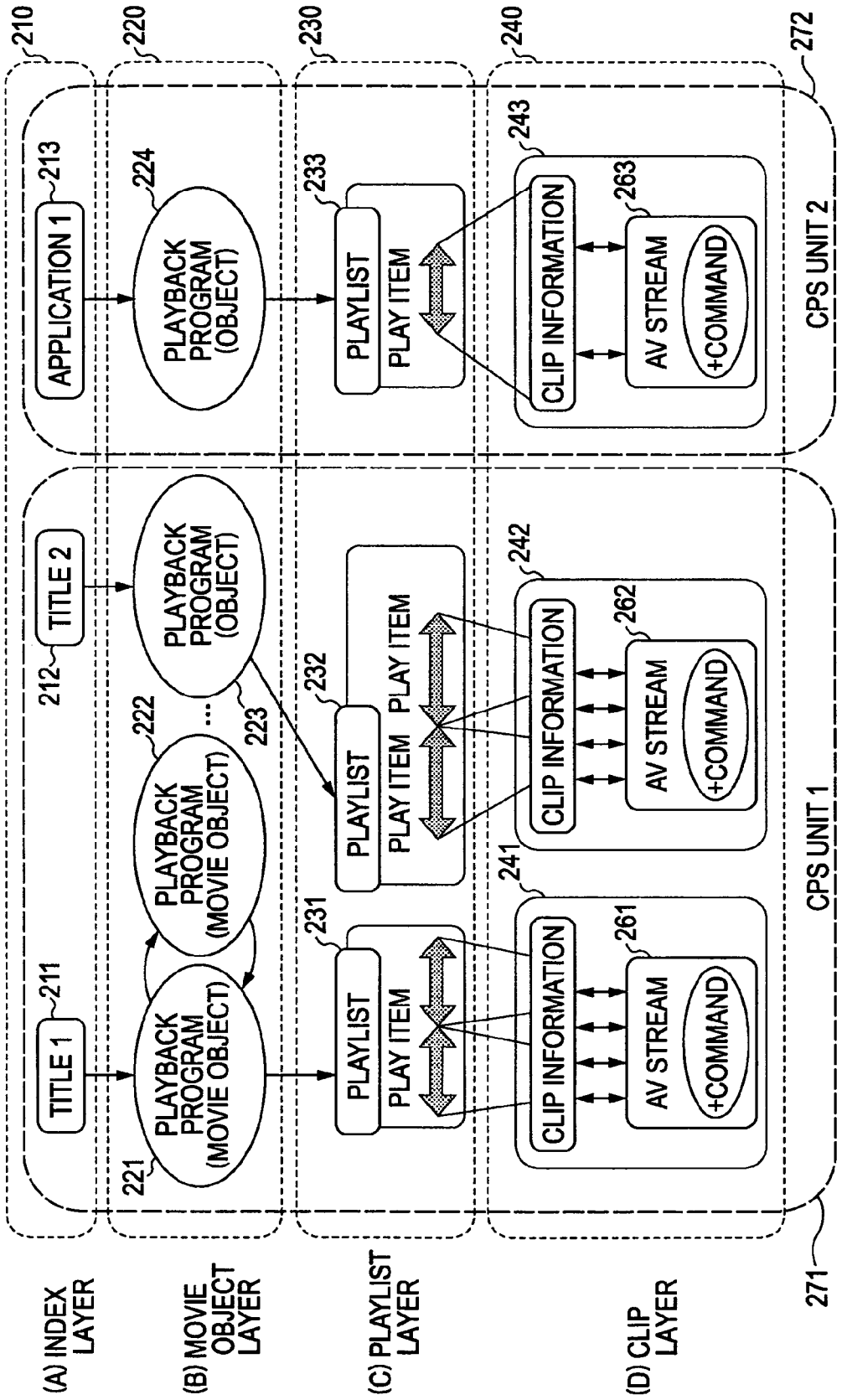

(A) INDEX LAYER
(B) MOVIE OBJECT LAYER
(C) PLAYLIST LAYER
(D) CLIP LAYER
210
220
230
240
271
272
CPS UNIT 1
CPS UNIT 2
TITLE 1
211
TITLE 2
212
APPLICATION 1
213
PLAYBACK PROGRAM (MOVIE OBJECT)
221
PLAYBACK PROGRAM (MOVIE OBJECT)
222
PLAYBACK PROGRAM (OBJECT)
223
PLAYBACK PROGRAM (OBJECT)
224
PLAYLIST
PLAY ITEM
231
PLAYLIST
PLAY ITEM PLAY ITEM
232
PLAYLIST
PLAY ITEM
233
CLIP INFORMATION
241
AV STREAM
+COMMAND
261
CLIP INFORMATION
242
AV STREAM
+COMMAND
262
CLIP INFORMATION
243
AV STREAM
+COMMAND
263

FIG. 3

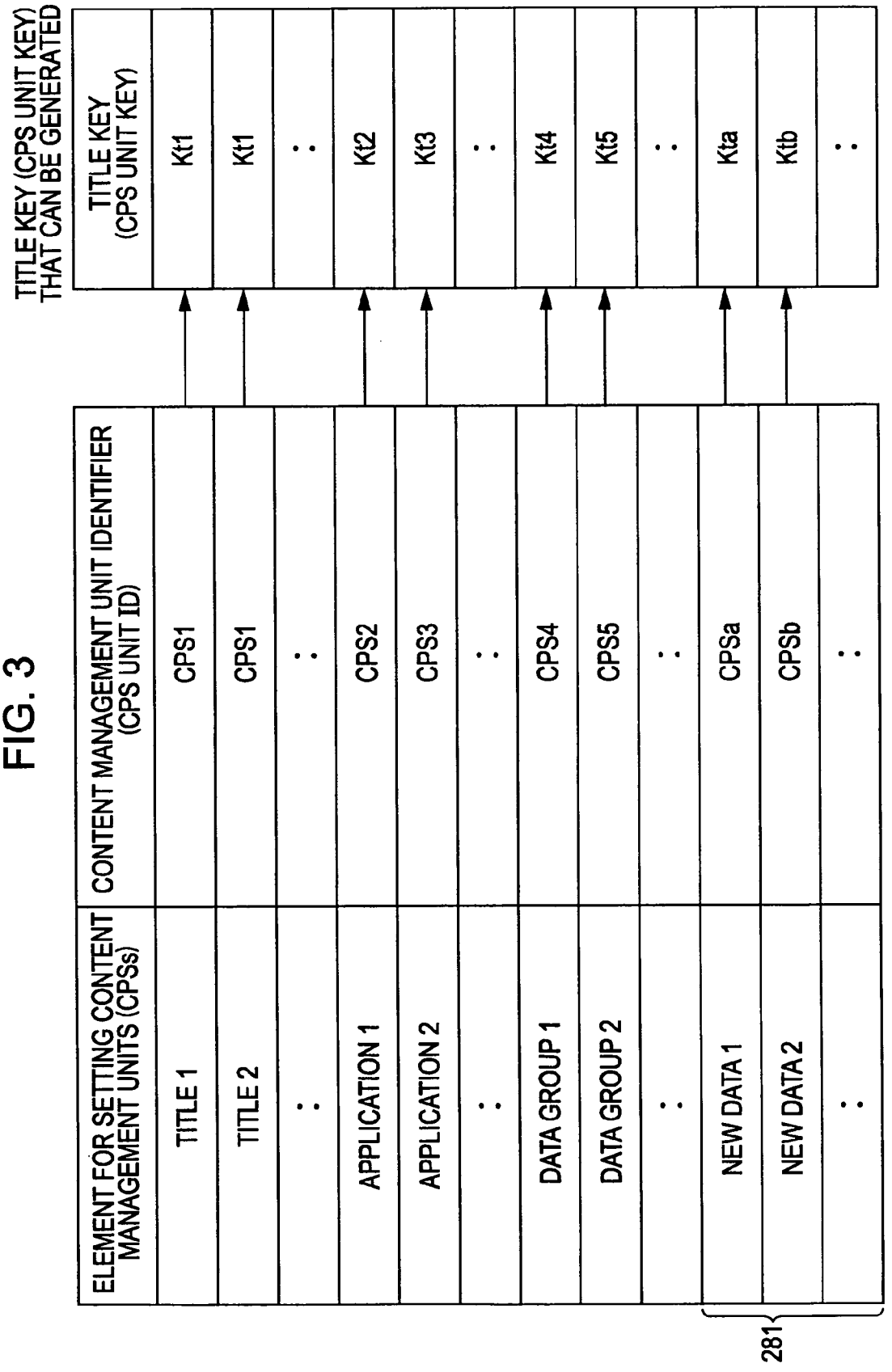

ELEMENT FOR SETTING CONTENT MANAGEMENT UNITS (CPSs)
CONTENT MANAGEMENT UNIT IDENTIFIER (CPS UNIT ID)
TITLE KEY (CPS UNIT KEY) THAT CAN BE GENERATED
TITLE KEY (CPS UNIT KEY)
TITLE 1
CPS1
Kt1
TITLE 2
CPS1
Kt1
APPLICATION 1
CPS2
Kt2
APPLICATION 2
CPS3
Kt3
DATA GROUP 1
CPS4
Kt4
DATA GROUP 2
CPS5
Kt5
NEW DATA 1
CPSa
Kta
NEW DATA 2
CPSb
Ktb
281

FIG. 4

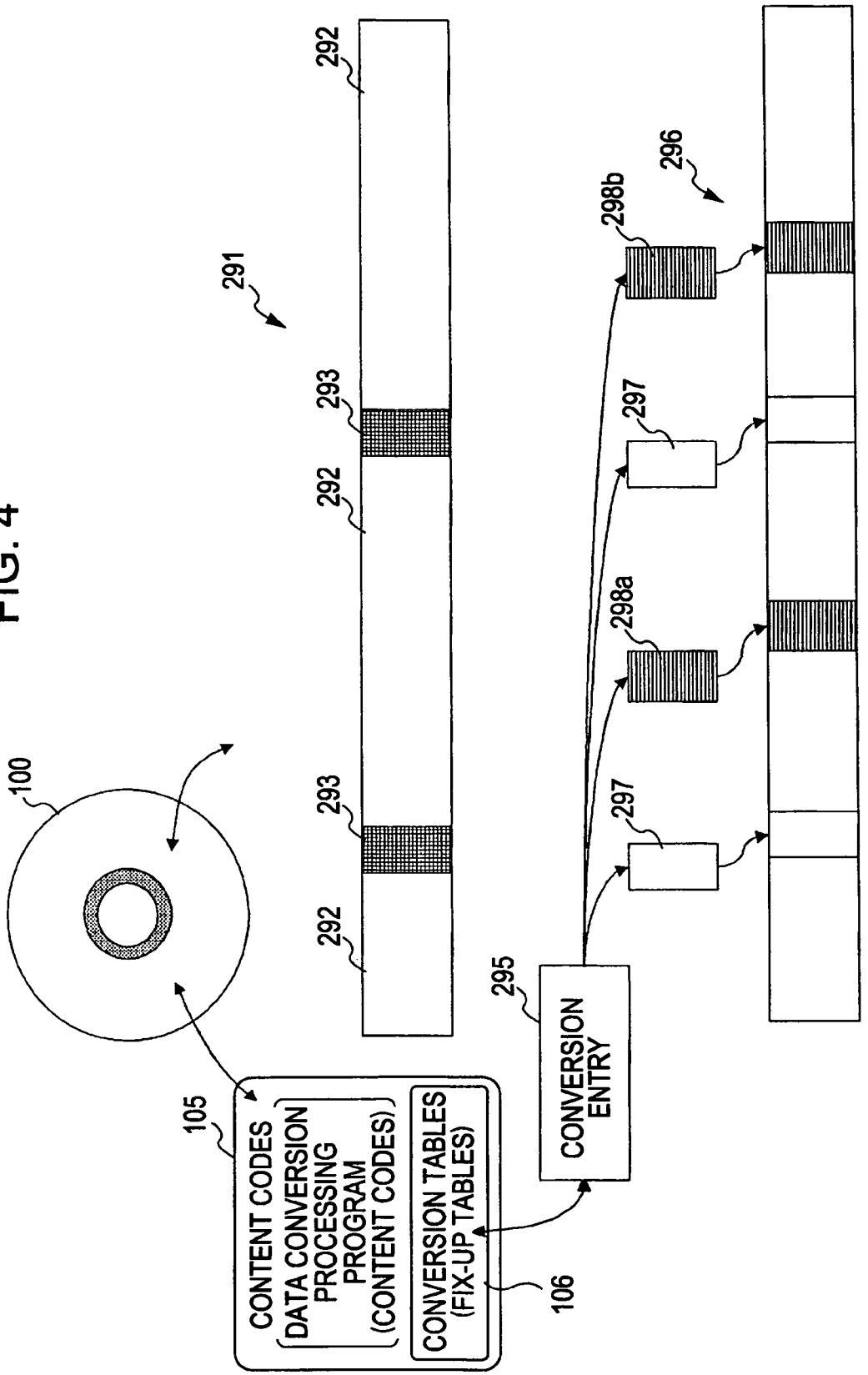

100
105
CONTENT CODES
DATA CONVERSION PROCESSING PROGRAM (CONTENT CODES)
CONVERSION TABLES (FIX-UP TABLES)
106
291
292
293
292
293
292
295
CONVERSION ENTRY
296
298b
297
298a
297

FIG. 5

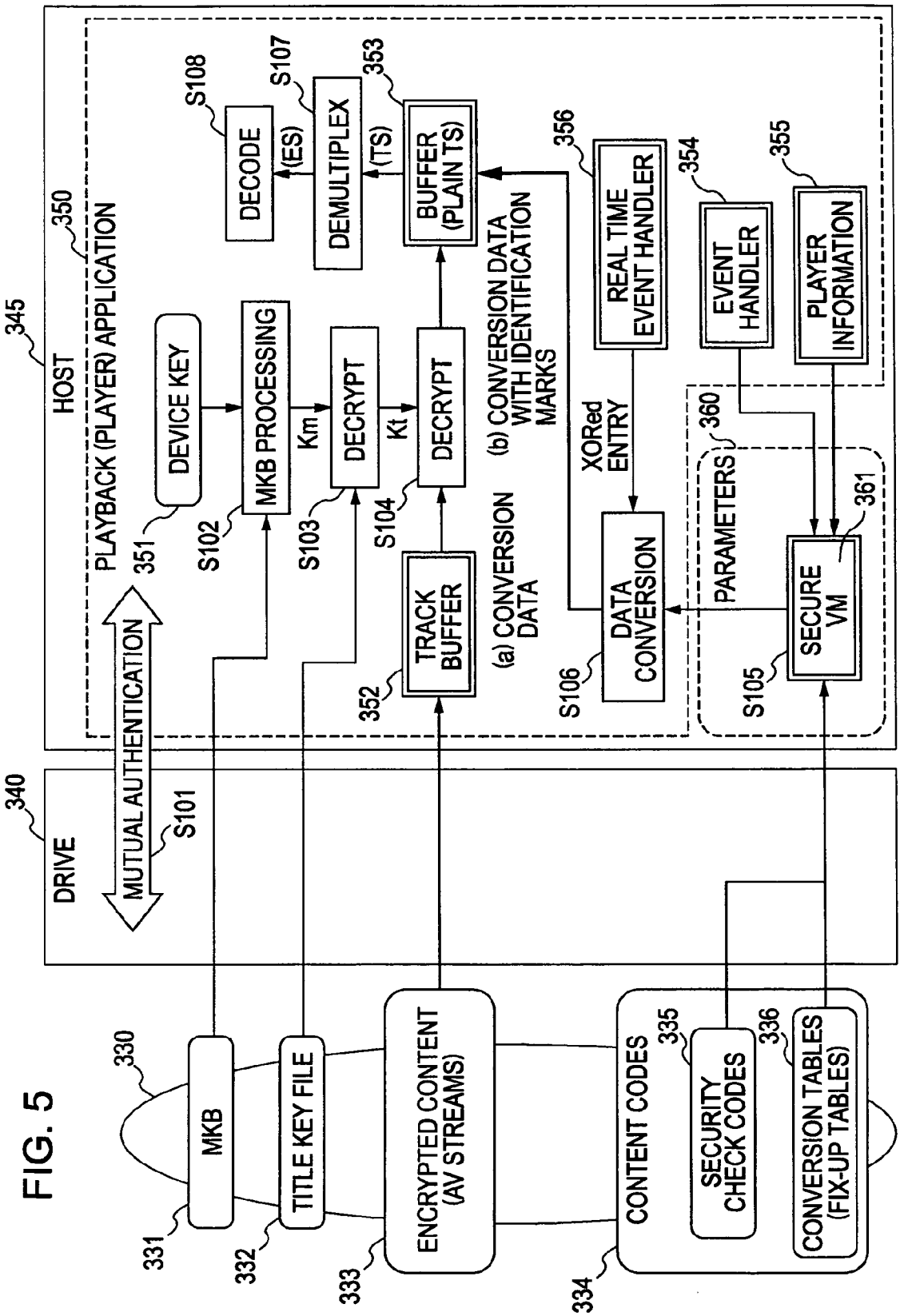

345
HOST
350
PLAYBACK (PLAYER) APPLICATION
340
DRIVE
MUTUAL AUTHENTICATION
S101
351
DEVICE KEY
S102
MKB PROCESSING
Km
S103
DECRYPT
Kt
S104
DECRYPT
352
TRACK BUFFER
(a) CONVERSION DATA
S108
DECODE
(ES)
S107
DEMULTIPLEX
(TS)
353
BUFFER (PLAIN TS)
(b) CONVERSION DATA WITH IDENTIFICATION MARKS
356
REAL TIME EVENT HANDLER
XORed ENTRY
S106
DATA CONVERSION
360
PARAMETERS
354
EVENT HANDLER
355
PLAYER INFORMATION
361
SECURE VM
S105
330
MKB
331
TITLE KEY FILE
332
ENCRYPTED CONTENT (AV STREAMS)
333
CONTENT CODES
334
335
SECURITY CHECK CODES
336
CONVERSION TABLES (FIX-UP TABLES)

FIG. 6

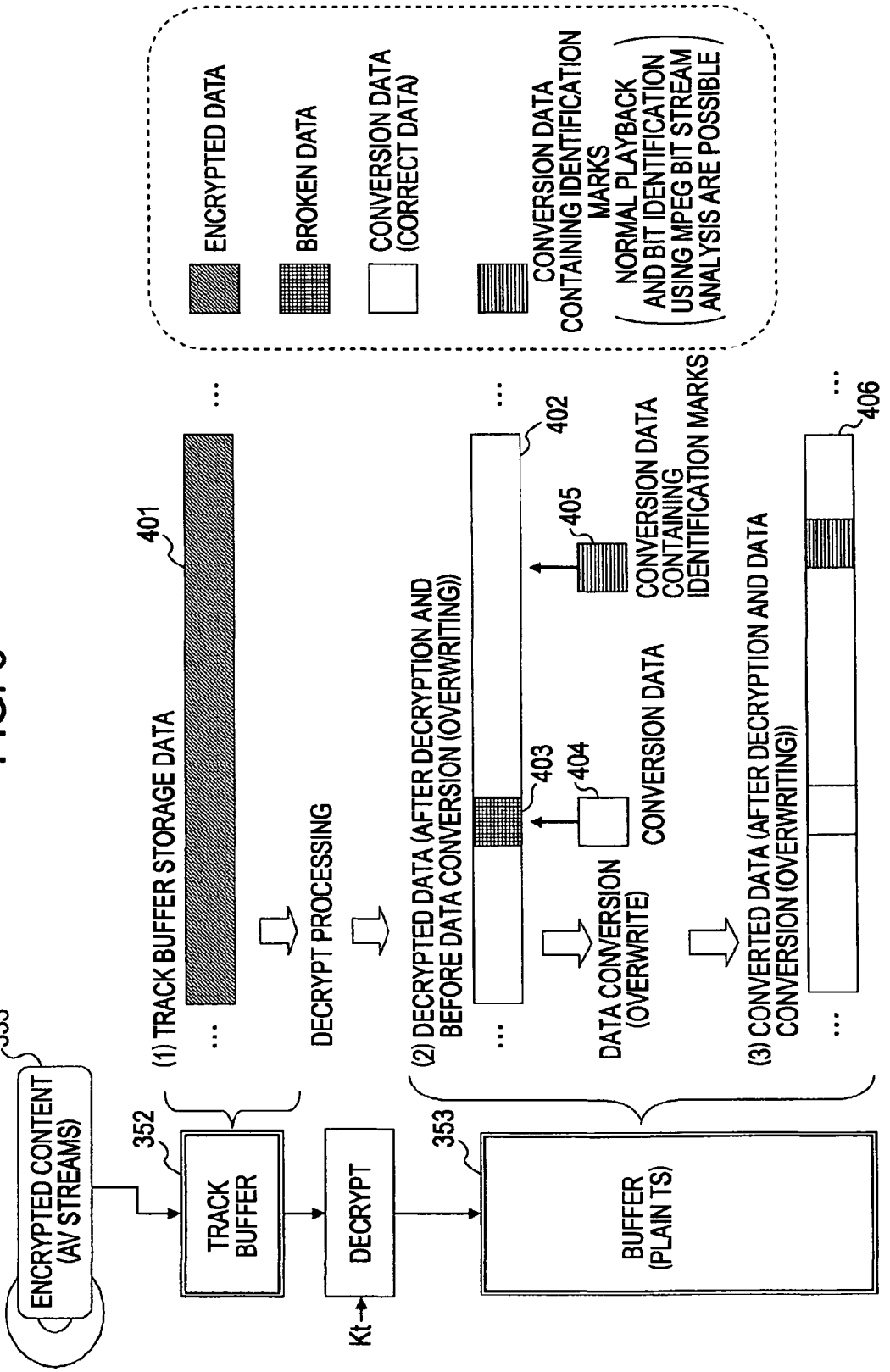

ENCRYPTED DATA
BROKEN DATA
CONVERSION DATA (CORRECT DATA)
CONVERSION DATA CONTAINING IDENTIFICATION MARKS
(NORMAL PLAYBACK AND BIT IDENTIFICATION USING MPEG BIT STREAM ANALYSIS ARE POSSIBLE)
333
ENCRYPTED CONTENT (AV STREAMS)
352
TRACK BUFFER
DECRYPT
Kt
353
BUFFER (PLAIN TS)
(1) TRACK BUFFER STORAGE DATA
401
DECRYPT PROCESSING
(2) DECRYPTED DATA (AFTER DECRYPTION AND BEFORE DATA CONVERSION (OVERWRITING))
402
403
404
405
DATA CONVERSION (OVERWRITE)
CONVERSION DATA
CONVERSION DATA CONTAINING IDENTIFICATION MARKS
(3) CONVERTED DATA (AFTER DECRYPTION AND DATA CONVERSION (OVERWRITING))
406

FIG. 7

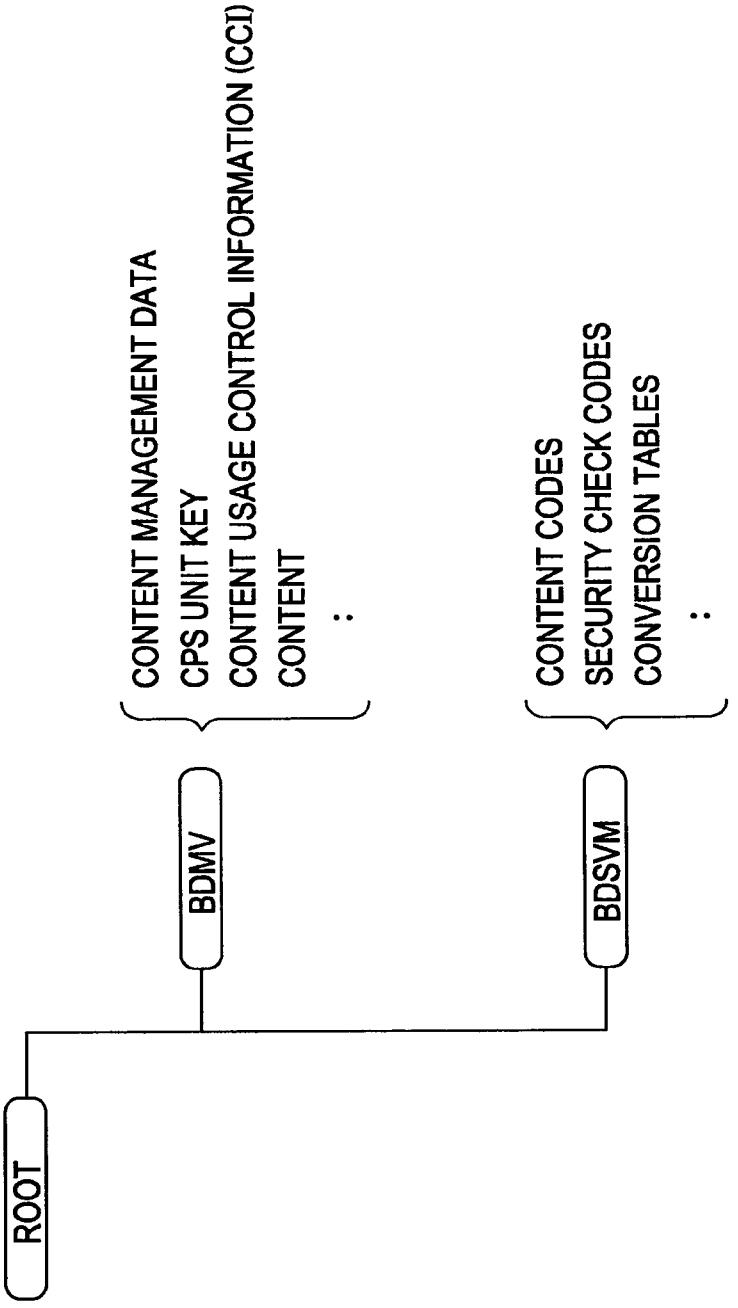

ROOT
BDMV
CONTENT MANAGEMENT DATA
CPS UNIT KEY
CONTENT USAGE CONTROL INFORMATION (CCI)
CONTENT
:
BDSVM
CONTENT CODES
SECURITY CHECK CODES
CONVERSION TABLES
:

FIG. 8

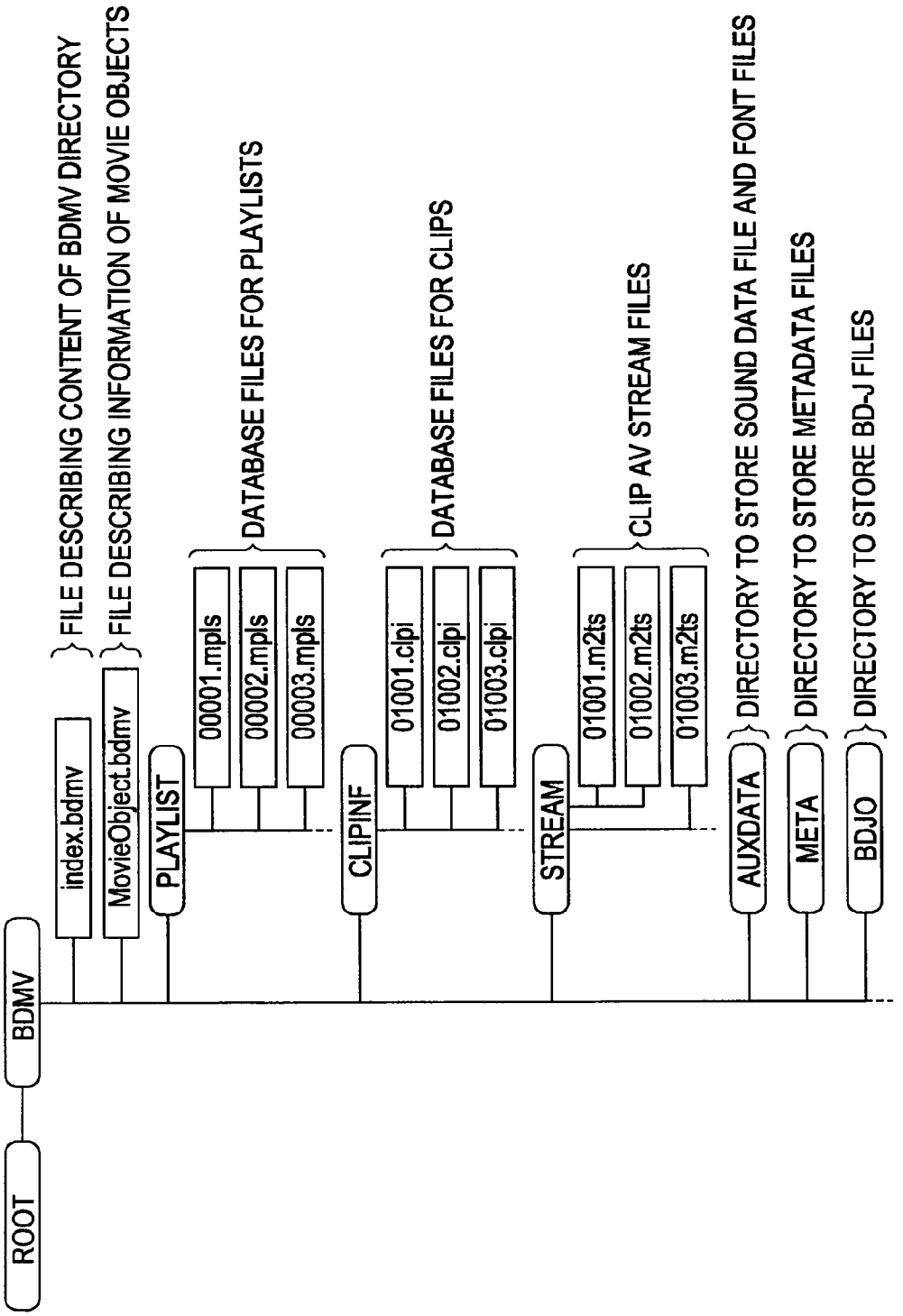

ROOT
BDMV
index.bdmv
FILE DESCRIBING CONTENT OF BDMV DIRECTORY
MovieObject.bdmv
FILE DESCRIBING INFORMATION OF MOVIE OBJECTS
PLAYLIST
00001.mpls
00002.mpls
00003.mpls
DATABASE FILES FOR PLAYLISTS
CLIPINF
01001.clpi
01002.clpi
01003.clpi
DATABASE FILES FOR CLIPS
STREAM
01001.m2ts
01002.m2ts
01003.m2ts
CLIP AV STREAM FILES
AUXDATA
DIRECTORY TO STORE SOUND DATA FILE AND FONT FILES
META
DIRECTORY TO STORE METADATA FILES
BDJO
DIRECTORY TO STORE BD-J FILES

FIG. 9

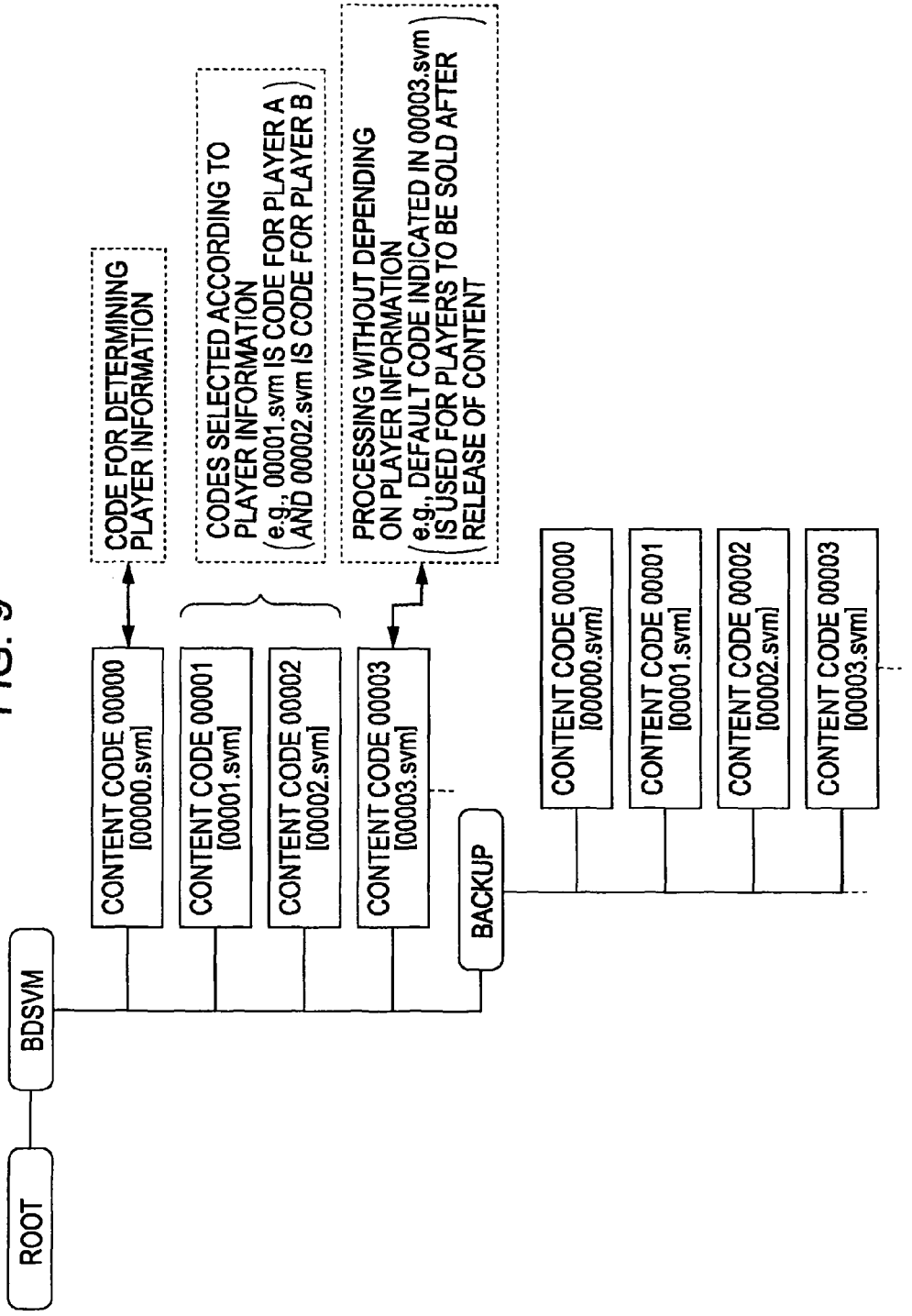

ROOT
BDSVM
CONTENT CODE 00000 [00000.svm]
CONTENT CODE 00001 [00001.svm]
CONTENT CODE 00002 [00002.svm]
CONTENT CODE 00003 [00003.svm]
BACKUP
CONTENT CODE 00000 [00000.svm]
CONTENT CODE 00001 [00001.svm]
CONTENT CODE 00002 [00002.svm]
CONTENT CODE 00003 [00003.svm]
CODE FOR DETERMINING PLAYER INFORMATION
CODES SELECTED ACCORDING TO PLAYER INFORMATION (e.g., 00001.svm IS CODE FOR PLAYER A AND 00002.svm IS CODE FOR PLAYER B)
PROCESSING WITHOUT DEPENDING ON PLAYER INFORMATION (e.g., DEFAULT CODE INDICATED IN 00003.svm IS USED FOR PLAYERS TO BE SOLD AFTER RELEASE OF CONTENT)

FIG. 10

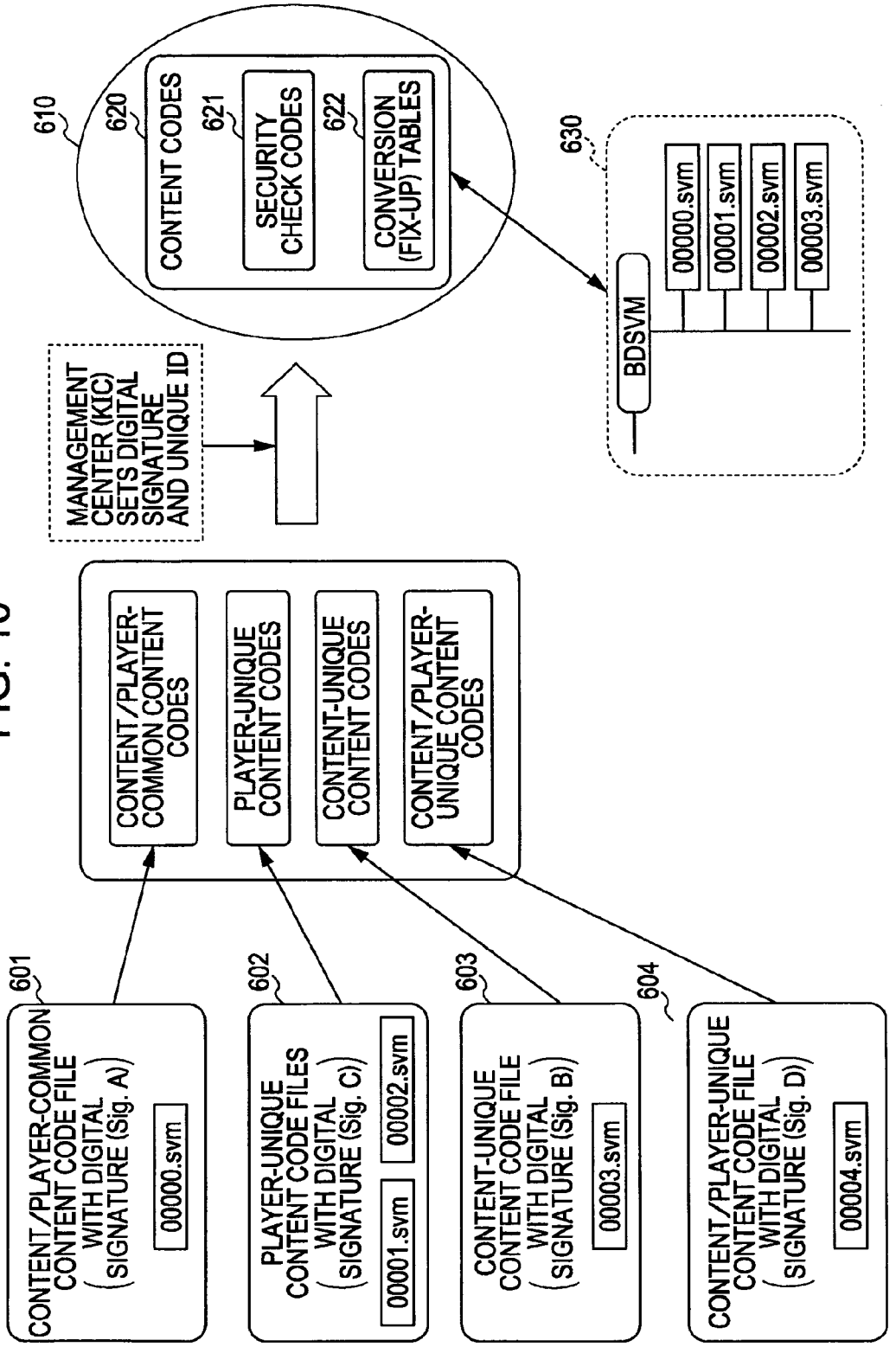

610
620
CONTENT CODES
621
SECURITY CHECK CODES
622
CONVERSION (FIX-UP) TABLES
630
BDSVM
00000.svm
00001.svm
00002.svm
00003.svm
MANAGEMENT CENTER (KIC) SETS DIGITAL SIGNATURE AND UNIQUE ID
CONTENT/PLAYER-COMMON CONTENT CODES
PLAYER-UNIQUE CONTENT CODES
CONTENT-UNIQUE CONTENT CODES
CONTENT/PLAYER-UNIQUE CONTENT CODES
601
CONTENT/PLAYER-COMMON CONTENT CODE FILE WITH DIGITAL SIGNATURE (Sig. A)
00000.svm
602
PLAYER-UNIQUE CONTENT CODE FILES WITH DIGITAL SIGNATURE (Sig. C)
00001.svm
00002.svm
603
CONTENT-UNIQUE CONTENT CODE FILE WITH DIGITAL SIGNATURE (Sig. B)
00003.svm
604
CONTENT/PLAYER-UNIQUE CONTENT CODE FILE WITH DIGITAL SIGNATURE (Sig. D)
00004.svm

FIG. 11

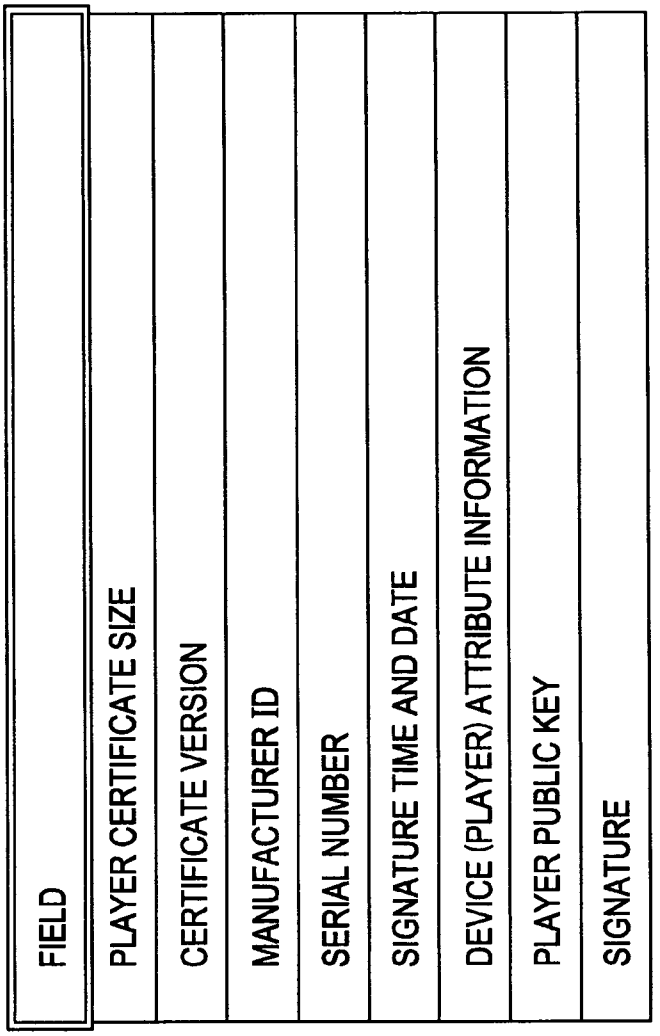

FIELD
PLAYER CERTIFICATE SIZE
CERTIFICATE VERSION
MANUFACTURER ID
SERIAL NUMBER
SIGNATURE TIME AND DATE
DEVICE (PLAYER) ATTRIBUTE INFORMATION
PLAYER PUBLIC KEY
SIGNATURE

FIG. 12

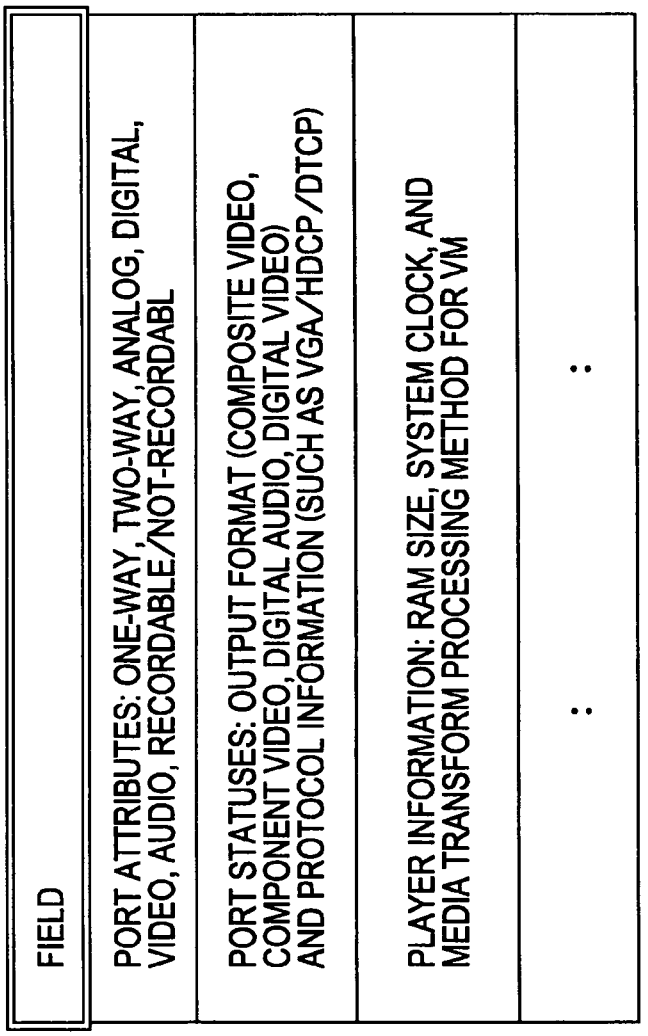

FIELD
PORT ATTRIBUTES: ONE-WAY, TWO-WAY, ANALOG, DIGITAL, VIDEO, AUDIO, RECORDABLE/NOT-RECORDABL
PORT STATUSES: OUTPUT FORMAT (COMPOSITE VIDEO, COMPONENT VIDEO, DIGITAL AUDIO, DIGITAL VIDEO) AND PROTOCOL INFORMATION (SUCH AS VGA/HDCP/DTCP)
PLAYER INFORMATION: RAM SIZE, SYSTEM CLOCK, AND MEDIA TRANSFORM PROCESSING METHOD FOR VM
:
:

FIG. 13

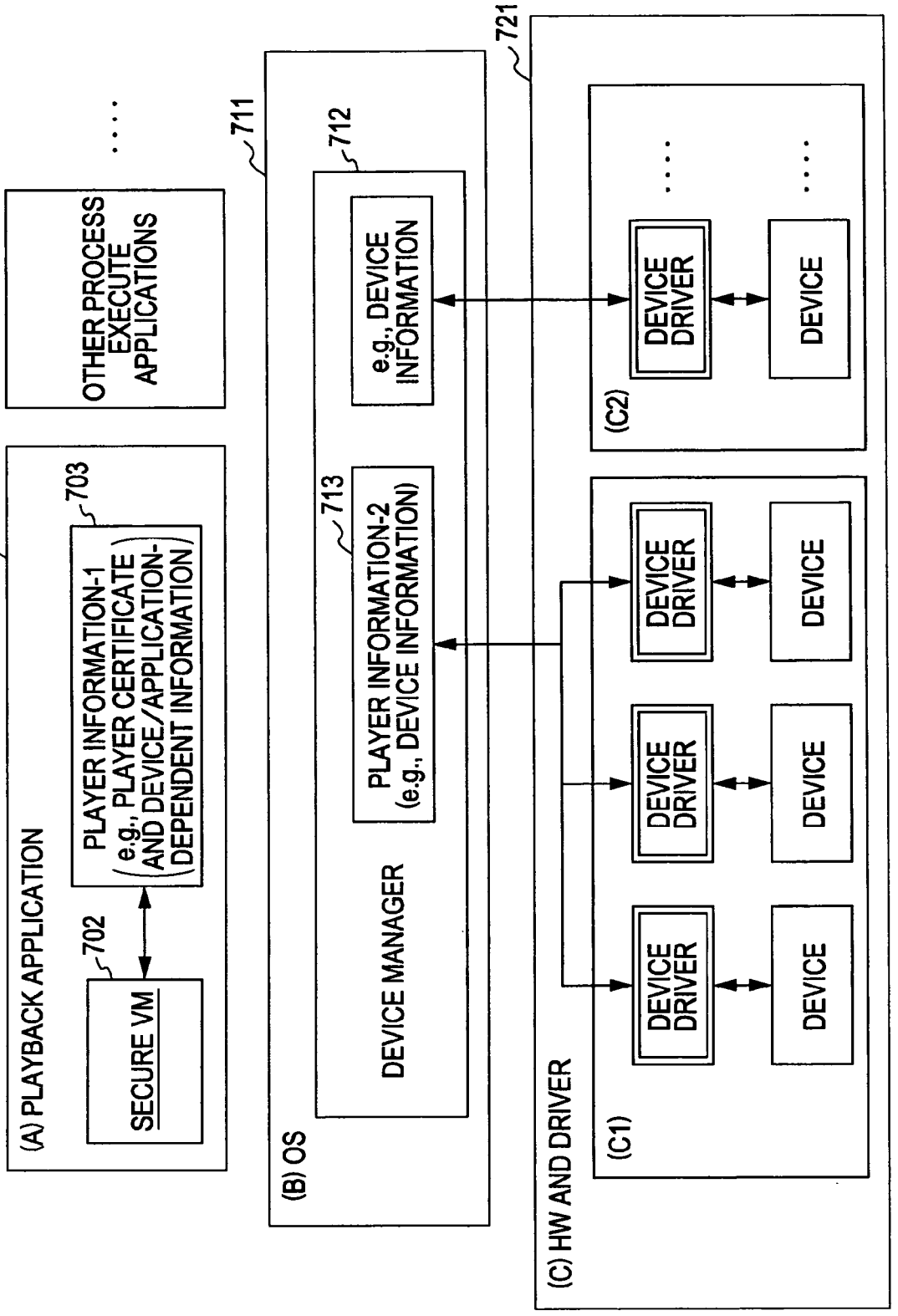

(A) PLAYBACK APPLICATION
701
SECURE VM
702
PLAYER INFORMATION-1
(e.g., PLAYER CERTIFICATE AND DEVICE/APPLICATION-DEPENDENT INFORMATION)
703
OTHER PROCESS EXECUTE APPLICATIONS
(B) OS
711
DEVICE MANAGER
PLAYER INFORMATION-2
(e.g., DEVICE INFORMATION)
713
e.g., DEVICE INFORMATION
712
(C) HW AND DRIVER
721
(C1)
DEVICE DRIVER
DEVICE
(C2)

FIG. 14

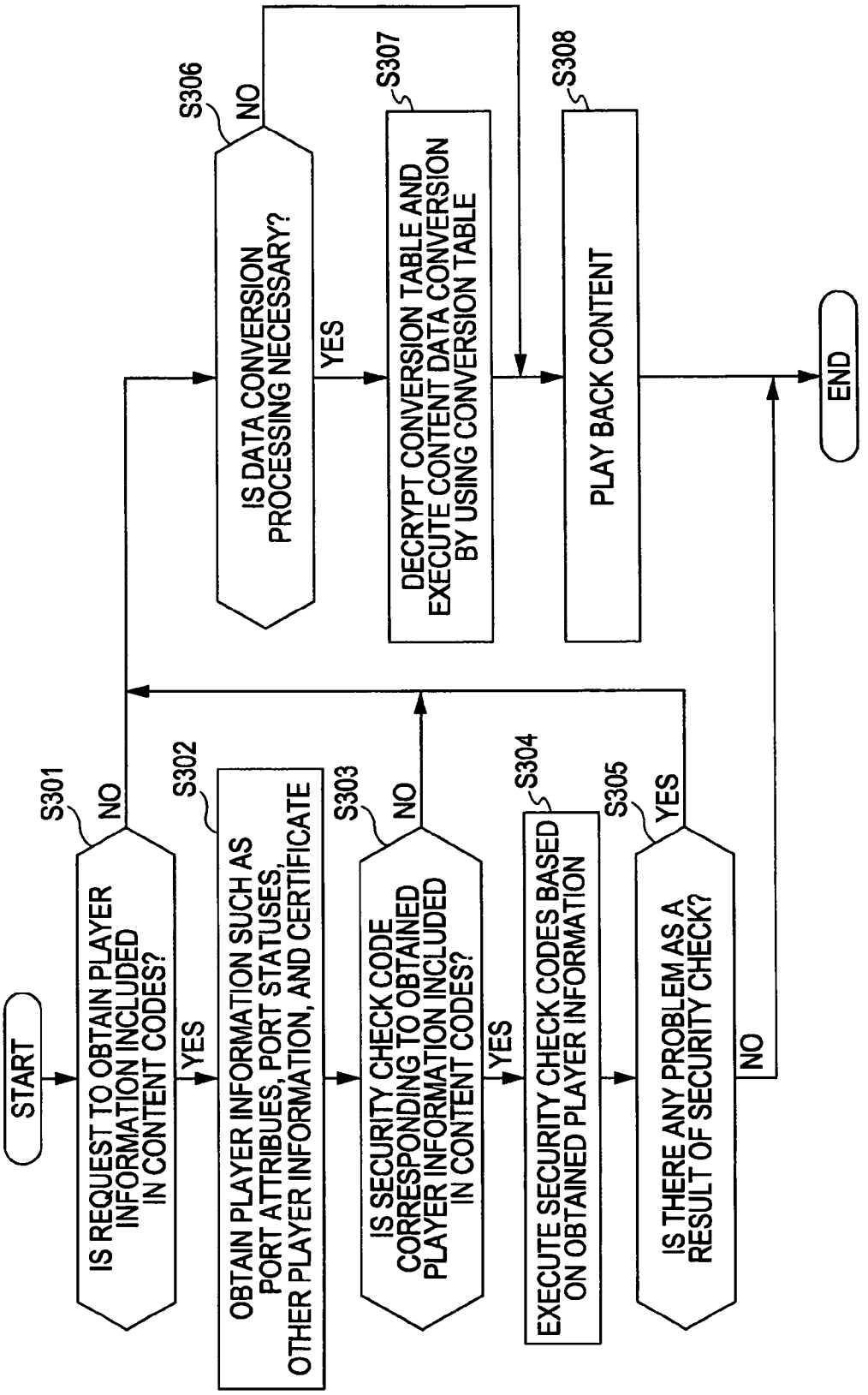

START
S301
IS REQUEST TO OBTAIN PLAYER INFORMATION INCLUDED IN CONTENT CODES?
NO
YES
S302
OBTAIN PLAYER INFORMATION SUCH AS PORT ATTRIBUES, PORT STATUSES, OTHER PLAYER INFORMATION, AND CERTIFICATE
S303
IS SECURITY CHECK CODE CORRESPONDING TO OBTAINED PLAYER INFORMATION INCLUDED IN CONTENT CODES?
NO
YES
S304
EXECUTE SECURITY CHECK CODES BASED ON OBTAINED PLAYER INFORMATION
S305
IS THERE ANY PROBLEM AS A RESULT OF SECURITY CHECK?
YES
NO
S306
IS DATA CONVERSION PROCESSING NECESSARY?
NO
YES
S307
DECRYPT CONVERSION TABLE AND EXECUTE CONTENT DATA CONVERSION BY USING CONVERSION TABLE
S308
PLAY BACK CONTENT
END

FIG. 15

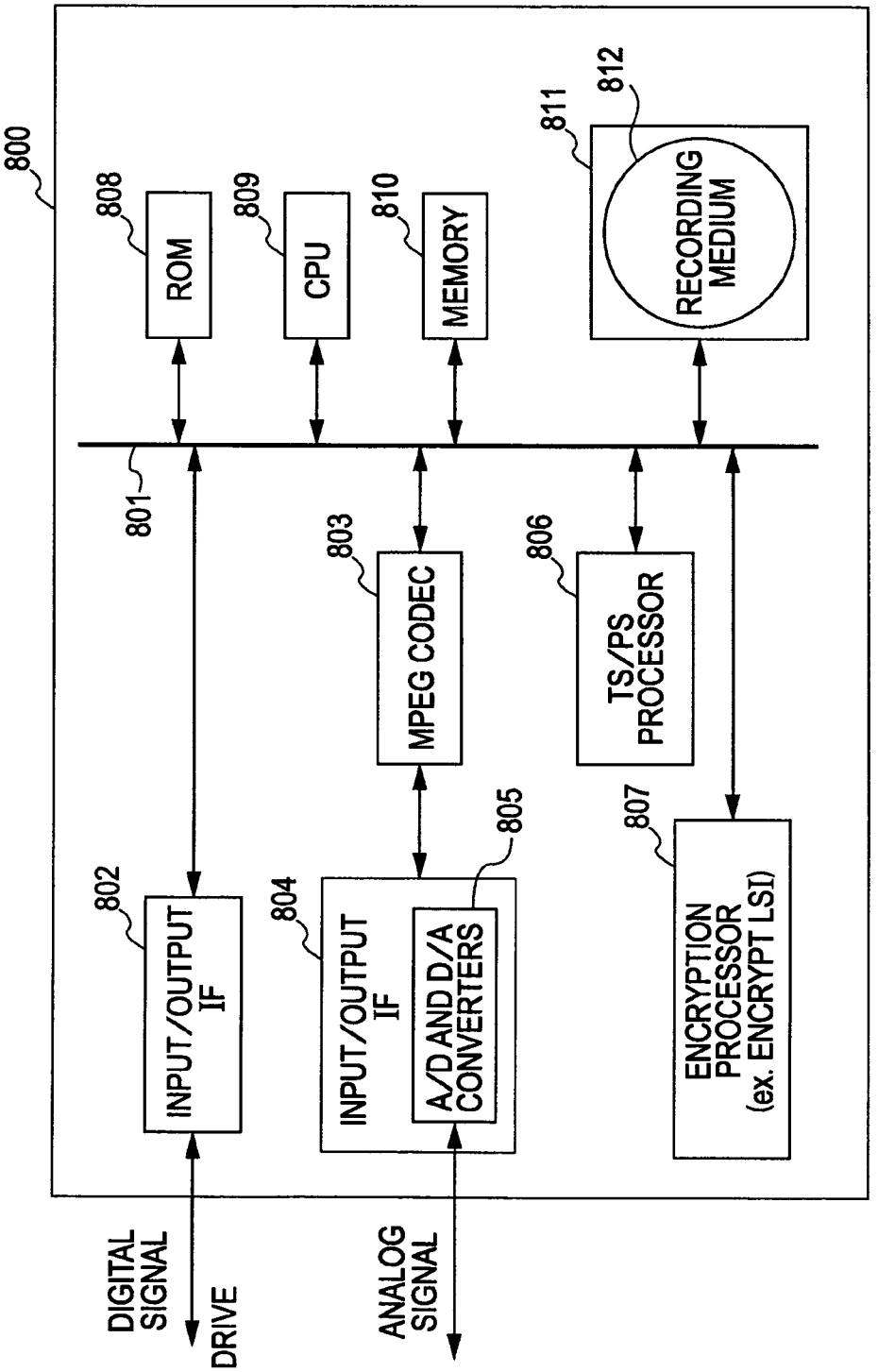
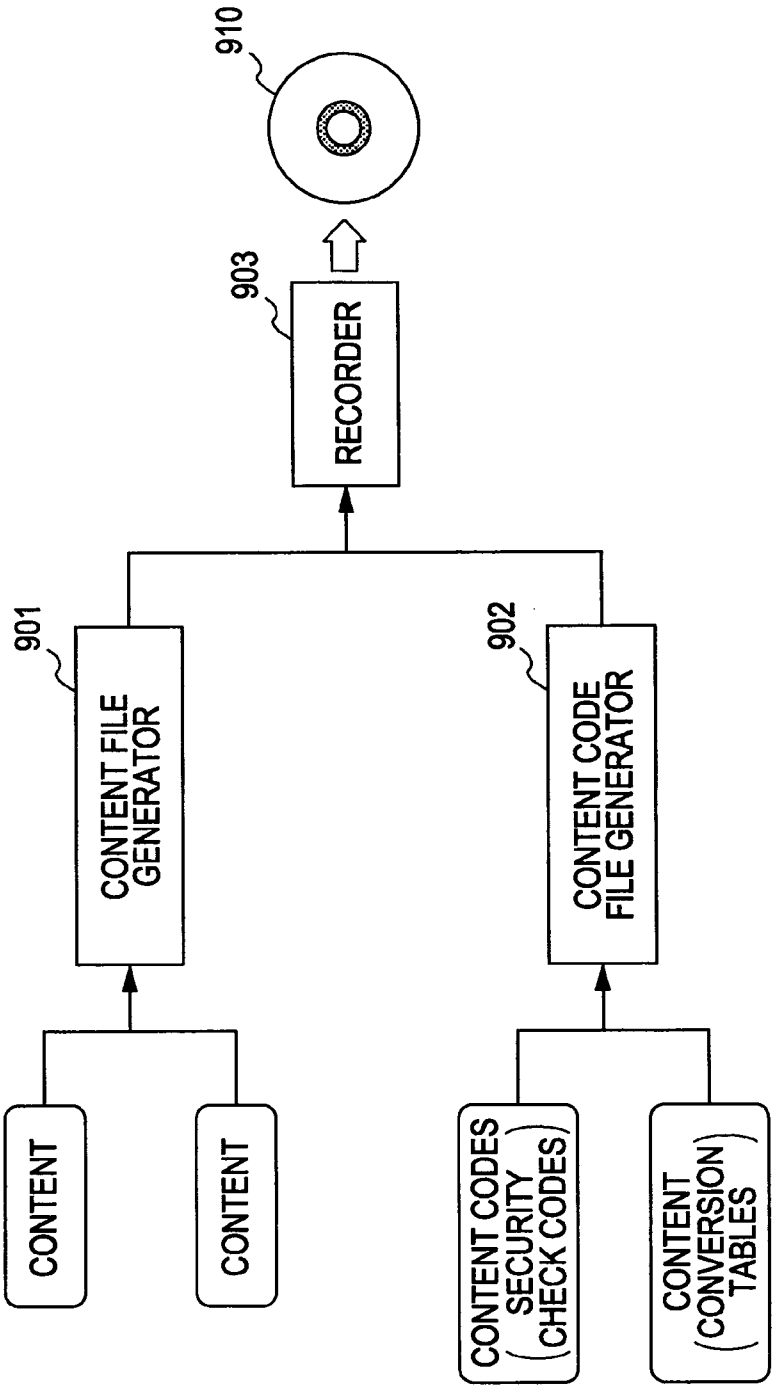

800
808
ROM
809
CPU
810
MEMORY
811
812
RECORDING MEDIUM
801
803
MPEG CODEC
806
TS/PS PROCESSOR
802
INPUT/OUTPUT IF
804
INPUT/OUTPUT IF
805
A/D AND D/A CONVERTERS
807
ENCRYPTION PROCESSOR (ex. ENCRYPT LSI)
DIGITAL SIGNAL
DRIVE
ANALOG SIGNAL

FIG. 16


910
903
RECORDER
901
CONTENT FILE GENERATOR
902
CONTENT CODE FILE GENERATOR
CONTENT
CONTENT
CONTENT CODES (SECURITY CHECK CODES)
CONTENT (CONVERSION TABLES)

INFORMATION PROCESSING APPARATUS AND METHOD, INFORMATION RECORDING MEDIUM MANUFACTURING APPARATUS AND METHOD, AND INFORMATION RECORDING MEDIUM

CROSS REFERENCES TO RELATED APPLICATIONS

The present invention contains subject matter related to Japanese Patent Application JP 2005-336226 filed in the Japanese Patent Office on Nov. 21, 2005, the entire contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

The present invention relates to information processing apparatuses and methods, information recording medium manufacturing apparatuses and methods, and information recording media. More particularly, the invention relates to an information processing apparatus and method, an information recording medium manufacturing apparatus and method, and an information recording medium that allow the execution of security check processing by using content codes recorded, as a content usage control program, on the information recording medium together with content.

2. Description of the Related Art

Various software data (hereinafter referred to as "content"), such as audio data, e.g., music, image data, e.g., movies, game programs, and various application programs, can be stored on recording media, for example, Blu-ray discs (trademark) using blue laser light, digital versatile discs (DVDs), mini discs (MDs), compact discs (CDs), as digital data. In particular, Blu-ray discs (trademark) using blue laser light are high-density recording discs and can record a large volume of video content as high-quality image data.

Digital content is stored in various information recording media, such as those described above, and is provided to users. A user uses the digital content by playing it back on the user's personal computer (PC) or a player, such as a disc player.

Generally, the distribution rights of many content data, such as music data and image data, are owned by creators or sellers of such content data. Basically, therefore, certain usage restrictions are imposed when distributing content, that is, the use of content is allowed only for authorized users, thereby preventing unauthorized copying.

According to digital recording apparatuses and recording media, images and sound can be repeatedly recorded and played back without a loss in quality. Thus, the distribution of illegally copied content via the Internet, the circulation of recording media, such as compact disc recordable (CD-R) discs, recording copied content thereon, so-called "pirated discs", and the use of copied content stored in hard disks of, for example, PCs, are widespread.

DVDs or large-capacity recording media, such as those using blue laser light which have recently been developed, can record a large volume of data, for example, up to several movies, on one medium, as digital information. Since video information can be recorded as digital information as described above, it is becoming important to perform copyright protection by preventing unauthorized copying. Nowadays, to prevent unauthorized copying of digital data, various techniques for preventing unauthorized copying are practically applied to digital recording apparatuses or recording media.

One of the measures to enhance copyright protection is to encrypt content. According to this measure, however, if leakage of cryptographic keys used for encrypting content occurs, it is difficult to prevent unauthorized copying of content and distribution of such copied content. One measure to solve such a problem is the configuration disclosed in Japanese Unexamined Patent Application Publication No. 2002-311998. In this configuration, content is recorded in the state in which part of content data is substituted by dummy data so that unauthorized playback of the content can be prevented.

When playing back content including dummy data, it is necessary to replace the dummy data by correct content data. This data conversion processing should be performed while preventing leakage of the correct content data, and certain measures are also preferably taken to prevent leakage of processing information, such as the location of dummy data and the conversion method.

As discussed above, when playing back content, it is necessary to execute content decryption processing and data conversion processing, and in some cases, security check, such as integrity checking processing to determine whether an information processing apparatus or a playback (player) program that is to use content is an authorized apparatus or program having a license, is conducted. Such data processing is performed by executing content codes recorded, as a content usage control program, on an information recording medium together with content. An example of content usage processing using content codes is disclosed in, for example, WO2005/008385.

Content codes are set as a file independent of content and are recorded on an information recording medium. Accordingly, it is possible that only content codes be moved or copied onto another information recording medium. If leakage of content codes occurs, which may lead to unauthorized distribution and usage of the content, it is likely that many items of content are illegally played back and used, causing enormous damage.

It can be assumed that various types of apparatuses or applications manufactured by different manufacturers be used for playing back content. If security check using content codes is performed according to only one common sequence, it is difficult to perform sufficient checking depending on the types of apparatuses or application. Additionally, a file for performing security check for each playback apparatus is probably used regardless of type of content, and thus, it is desirable that the file be reused.

SUMMARY OF THE INVENTION

It is thus desirable to provide an information processing apparatus and method, an information recording medium manufacturing apparatus and method, and an information recording medium in which a strict management structure for content codes recorded, as a content usage control program, on an information recording medium together with content can be implemented while allowing the reuse of security check codes included in the content codes, and when verifying players by using the security check codes, optimal security check in accordance with the types of players can be conducted.

According to an embodiment of the present invention, there is provided an information processing apparatus that performs playback processing for content recorded on an information recording medium. The information processing apparatus includes a data processor operable to obtain a security check code included in content codes recorded on the information recording medium when using the content and to

conduct security check for the information processing apparatus in accordance with the obtained security check code. The data processor obtains information based on at least one of identification information and configuration information corresponding to the information processing apparatus or a content usage application, and among security check codes classified into a plurality of data files, the data processor selects a security check code corresponding to the obtained information to conduct security check on the basis of the selected security check code.

The data processor may obtain a player certificate stored in a storage unit of the information processing apparatus to verify the integrity of the player certificate, and under the condition that the integrity of the player certificate is verified, the data processor may obtain at least one of the identification information and the configuration information corresponding to the information processing apparatus or the content usage application from information recorded on the player certificate.

The data processor may obtain the information based on, as the configuration information, at least one of port attribute information, port status information, memory area size information, and data processing mode information concerning the information processing apparatus, and conducts security check corresponding to the obtained information.

When obtaining the configuration information, the data processor may obtain device information managed by an operating system (OS) and conducts security check corresponding to the obtained information.

When the configuration information is updated, the data processor may obtain the updated configuration information and conducts security check corresponding to the updated configuration information.

The data processor may perform data processing which accompanies content data replacement processing using a conversion table included in the content codes, the data processing being performed in accordance with a selected code corresponding to the information obtained based on at least one of the identification information and the configuration information corresponding to the information processing apparatus or the content usage application.

According to another embodiment of the present invention, there is provided an information recording medium manufacturing apparatus including content file obtaining means for obtaining content files storing content data to be recorded on an information recording medium, content code file obtaining means for obtaining content code files storing content codes including a security check processing program to be executed when content is used, and recording means for recording the content files obtained by the content file obtaining means and the content code files obtained by the content code file obtaining means on the information recording medium. The content code file obtaining means obtains a plurality of content code files corresponding to types of information processing apparatuses or content usage applications.

The content code file obtaining means may obtain content code files storing common content codes that do not depend on the types of information processing apparatuses or content usage applications and content code files storing content codes that depend on the types of information processing apparatuses or content usage applications.

The content code file obtaining means may obtain a code file storing a code for determining the type of information processing apparatus or content usage application.

According to another embodiment of the present invention, there is provided an information recording medium including content files storing content data and content code files storing content codes including a security check processing program to be executed when content is used. The content code files include a plurality of content code files corresponding to types of information processing apparatuses and content usage applications.

The content code files may include content code files storing common content codes that do not depend on the types of information processing apparatuses or content usage applications and content code files storing content codes that depend on the types of information processing apparatuses or content usage applications.

The content code files may include a code file storing a code for determining the type of information processing apparatus or content usage application.

According to another embodiment of the present invention, there is provided an information processing method, for use in an information processing apparatus, for performing data processing using data recorded on an information recording medium. The information processing method includes the steps of performing data processing by obtaining a security check code included in content codes recorded on the information recording medium when using content recorded on the information recording medium and by conducting security check for the information processing apparatus in accordance with the obtained security check code. In the execution of the data processing, information based on at least one of identification information and configuration information corresponding to the information processing apparatus or a content usage application is obtained, and among security check codes classified into a plurality of data files, a security check code corresponding to the obtained information is selected, and security check is conducted on the basis of the selected security check code.

In the execution of the data processing, a player certificate stored in a storage unit of the information processing apparatus may be obtained to verify the integrity of the player certificate, and under the condition that the integrity of the player certificate is verified, at least one of the identification information and the configuration information corresponding to the information processing apparatus or the content usage application may be obtained from information recorded on the player certificate.

In the execution of the data processing, at least one of port attribute information, port status information, memory area size information, and data processing mode information concerning the information processing apparatus may be obtained as the configuration information, and security check corresponding to the obtained information is conducted.

In the execution of the data processing, device information managed by an OS may be obtained when obtaining the configuration information, and security check corresponding to the obtained information is conducted.

In the execution of the data processing, when the configuration information is updated, the updated configuration information may be obtained, and security check corresponding to the updated configuration information is conducted.

In the execution of the data processing, data processing which accompanies content data replacement processing using a conversion table included in the content codes may be performed, the data processing being performed in accordance with a selected code corresponding to the information obtained based on at least one of the identification information and the configuration information corresponding to the information processing apparatus or the content usage application.

According to another embodiment of the present invention, there is provided an information recording medium manufac-

turing method for use in an information recording medium manufacturing apparatus. The information recording medium manufacturing method includes the steps of obtaining content files storing content data to be recorded on an information recording medium, obtaining content code files storing content codes including a security check processing program to be executed when content is used, and recording the obtained content files and the obtained content code files on the information recording medium. In the execution of obtaining the content code files, a plurality of content code files corresponding to types of information processing apparatuses or content usage applications are obtained.

In the execution of obtaining the content code files, content code files storing common content codes that do not depend on the types of information processing apparatuses or content usage applications and content code files storing content codes that depend on the types of information processing apparatuses or content usage applications may be obtained.

In the execution of obtaining the content code files, a code file storing a code for determining the type of information processing apparatus or content usage application may be obtained.

According to an embodiment of the present invention, a security check code included in content codes recorded on an information recording medium is obtained when using the content, and security check for an information processing apparatus is conducted in accordance with the obtained security check code. Information based on at least one of identification information and configuration information corresponding to the information processing apparatus or a content usage application is obtained, and a security check code corresponding to the obtained information is selected, and security check is conducted on the basis of the selected security check code. Accordingly, optimal security check in accordance with various types and versions of apparatuses or various playback applications can be conducted, thereby effectively preventing unauthorized use of content. Additionally, the reuse of security codes can be implemented, thereby achieving efficient manufacturing of information recording media.

Further features and advantages of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** illustrates storage data in an information recording medium and the configuration and processing of a drive and an information processing apparatus;

FIG. **2** illustrates content management units set in storage content in an information recording medium;

FIG. **3** illustrates the association between content management units set in storage content in an information recording medium and unit keys assigned to the content management units;

FIG. **4** illustrates content recorded on an information recording medium and data conversion processing necessary for content playback;

FIG. **5** illustrates an example of content playback processing;

FIG. **6** illustrates data conversion processing executed when content is played back;

FIG. **7** illustrates the directory structure of data to be recorded on an information recording medium;

FIG. **8** illustrates the directory structure of content and management data to be recorded on an information recording medium;

FIG. **9** illustrates the directory structure of content codes to be recorded on an information recording medium;

FIG. **10** illustrates details of a generation/recording process of content codes to be recorded on an information recording medium;

FIG. **11** illustrates an example of the data structure of a player certificate;

FIG. **12** illustrates player information other than a player certificate;

FIG. **13** illustrates an example of processing for obtaining player information;

FIG. **14** is a flowchart illustrating a content playback sequence performed by an information processing apparatus;

FIG. **15** illustrates an example of the hardware configuration of an information processing apparatus; and

FIG. **16** is a block diagram illustrating the configuration of an information recording medium manufacturing apparatus.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

Details of an information processing apparatus and method, an information recording medium manufacturing apparatus and method, and an information recording medium according to an embodiment of the present invention are described below with reference to the accompanying drawings. A description thereof is given in the following order.

1. Overview of Storage Data in Information Recording Medium and Processing Performed by Drive and Host
2. Content Management Units (CPS Units)
3. Overview of Structure of Content Data Including Modified Data and Data Conversion Processing
4. Content Playback Processing
5. Processing Using Security Check Codes
6. Configuration of Information Processing Apparatus
7. Information Recording Medium Manufacturing Apparatus and Information Recording Medium

1. Overview of Storage Data in Information Recording Medium and Processing Performed by Drive and Host

An overview of storage data in an information recording medium and processing performed by a drive and a host is first described. FIG. **1** illustrates the configurations of an information recording medium **100** in which content is stored, a drive **120**, and a host **140**. The host **140** is a data playback (or recording) application executed by an information processing apparatus, for example, a PC, and executes processing by using hardware of an information processing apparatus, for example, a PC, in accordance with a predetermined data processing sequence.

The information recording medium **100** is an information recording medium (e.g., a read only memory (ROM) disk), or a data rewritable information recording medium (e.g., an RE disk), such as a Blu-ray disc (trademark) or a DVD, storing authorized content, manufactured in a disc manufacturing factory with the permission of a so-called "content right holder" having an authorized content copyright or distribution right. In the following embodiment, as the information recording medium, a disc medium is used. In the present invention, however, various forms of information recording media can be used.

The information recording medium **100** stores, as shown in FIG. **1**, encrypted content **101** subjected to encrypt processing and replacement (overwriting) processing, i.e., in which part of data is overwritten by another data, a media key block

(MKB) **102**, which serves as a cryptographic key block, generated based on a tree-structure key distribution system, which is known as one mode of a broadcast encryption system, a title key file **103** including an encrypted title key (encrypted content protection system (CPS) unit key) used for content decryption processing, usage rule information **104** including copy control information (CCI) as content copy/playback control information, and content codes **105** including a data conversion processing program executed when using the encrypted content **101**.

The content codes **105** include conversion tables (fix-up tables) **106** storing conversion data corresponding to data to be replaced in predetermined areas of content, and also include security check codes **107**, such as a program for verifying the integrity of a player (playback apparatus) that performs content playback.

When playing back content, the information processing apparatus verifies, for example, the integrity of the player in accordance with the security check codes **107** included in the content codes **105**, and then extracts conversion data stored in the conversion table (fix-up table) **106** included in the content codes **105** in accordance with the data conversion processing program contained in the content codes **105** to replace part of data forming the content.

The conversion tables **106** and the security check codes **107** include conversion tables and various types of codes so that processing, i.e., security check and conversion processing, can be performed in accordance with the types of players and player port information. For example, the conversion tables **106** and the security check codes **107** include a conversion table and a security check code corresponding to players manufactured by company A and a conversion table and a security check code corresponding to players manufactured by company B. To use content, a player selects the security check code and the conversion table corresponding to that player and executes processing.

The content codes **105** include, not only the data conversion processing program, but also information and programs for executing various other types of processing, such as startup processing and security check processing. Details of the content codes **105** are given below. The storage data in the information recording medium **100** shown in FIG. **1** are examples only, and different storage data can be stored depending on the type of disc. An overview of various items of information is discussed below.

Encrypted Content

In the information recording medium **100**, various items of content data, such as audiovisual (AV) streams of moving picture content, for example, high definition (HD) movie content, game programs, image files, sound data, and text data, defined by specific standards, are stored. Those items of content are specific AV format standard data, which are stored according to specific AV data formats. More specifically, for example, content is stored as the Blu-ray disc (trademark) ROM standard data according to the Blu-ray ROM standard format.

Game programs, image files, sound data, or text data, which serve as service data, may be stored. Those items of content may be stored as data having a data format which is not compliant with a specific AV data format.

As the types of content, various content, such as music data, image data, for example, moving pictures and still images, game programs, and WEB content, are included. Such content includes various modes of information, such as content information that can be used only by data from the information recording medium **100**, and content information that can be used by a combination of the data from the information recording medium **100** and data provided from a server connected to the recording medium **100** via a network.

To individually control the use of each segment of content, the content is stored in the information recording medium **100** by assigning different keys (CPS unit keys or unit keys, which are also referred to as "title keys") to segments and by encrypting each segment with a key, which is different from those for the other segments. The unit to which one unit key is assigned is referred to as a "content management unit (CPS unit)". In the content, part of the data is set as broken data by being replaced by data different from correct data, and thus, even if the content is decrypted, it cannot be correctly played back. To correctly play back the content, it is necessary to replace broken data by correct data registered in the conversion table. Such replacement processing is described in detail below.

MKB

The MKB **102** is a cryptographic key block generated based on a tree-structure key distribution system, which is known as one mode of the broadcast encryption method. The MKB **102** is a key information block that makes it possible to obtain a media key (Km), which is a key necessary for decrypting content, only by processing (decryption) by using a device key (Kd) stored in information processing apparatuses having valid licenses. The MKB **102** is based on an information distribution system according to a so-called hierarchical tree structure. The MKB **102** makes it possible to obtain the media key (Km) only when a user device (information processing apparatus) has a valid license, and to disable revoked user devices from obtaining the media key (Km).

By changing the device key used for encrypting key information stored in the MKB **102**, a management center, which serves as a license entity, allows the MKB **102** to disable a device key stored in a specific user device from decrypting content, that is, makes the user device unable to obtain the media key necessary for decrypting the content. It is thus possible to provide encrypted content only to devices having valid licenses while revoking unauthorized devices at a suitable time. Content decrypt processing is discussed below.

Title Key File

As stated above, in each content or a set of a plurality of items of content, content segments are encrypted with individual cryptographic keys (title keys or CPS unit keys) and are then stored in the information recording medium **100**. That is, it is necessary to divide AV streams, music data, image data, such as moving pictures and still images, game programs, and WEB content, into units, which are management units for using content, and to generate different title keys for the divided units and to decrypt the content units with the generated different title keys. Information for generating the title keys is title key data, and title keys can be obtained by decrypting encrypted title keys with keys generated by using, for example, media keys. Title keys for the corresponding units are generated in accordance with a predetermined cryptographic key generation sequence using the title key data so that the content units can be decrypted.

Usage Rule Information

Usage rule information includes, for example, copy/playback control information (CCI), that is, copy restriction information or playback restriction information used for controlling the use of the encrypted content **101** stored in the information recording medium **100**. The copy/playback control information (CCI) may be set in various manners, such as being set for each CPS unit, which serves as the content management unit, or for a plurality of CPS units.

Content Codes

As stated above, the content codes **105** include the conversion tables (fix-up tables) **106** in which conversion data corresponding to data to be replaced in predetermined areas of content is registered and the security check codes **107**, such as a program for verifying the integrity of players (playback apparatuses) that perform content playback.

As discussed above, the conversion tables **106** and the security check codes **107** include various codes so that processing can be performed in accordance with the types of various players and player port information. To use content, a player selects the security check code and the conversion table corresponding to that player and executes security check processing and data conversion processing, respectively.

The host **140**, which serves as the playback application **150** of a player to play back content, sets a virtual machine (VM) that executes data conversion processing, and executes security check processing and data conversion processing in accordance with the content codes read from the information recording medium **100** by using the VM to convert part of the data forming the content by using the entries registered in the conversion tables **106**.

The encrypted content **101** stored in the information recording medium **100** has been subjected to certain encryption processing, and part of the data forming the content **101** is broken data different from correct data. When playing back the encrypted content **101**, it is necessary to replace (overwrite) a plurality of broken data by conversion data registered in the conversion tables **106**. Many broken data are dispersedly set in the encrypted content **101**, and to play back the content **101**, it is necessary to replace (overwrite) the plurality of broken data by the conversion data registered in the conversion tables **106**. Because of the existence of broken data, even if leakage of cryptographic keys occurs and the unauthorized decryption of content is performed, the content cannot be played back correctly. As a result, unauthorized use of the content can be prevented.

The conversion tables **106** include, not only normal conversion data, but also specific conversion data that enables the analysis of bits forming identification information for identifying a content player or a content playback application. More specifically, the specific conversion data includes identification marks on which player IDs as the identification data for players (devices that execute host applications) or identification information generated based on the player IDs are recorded. The specific conversion data including identification marks are generated by slightly modifying the bits forming the correct content data to such a degree as not to influence playback of the content.

The content codes **105** include, not only the data conversion processing program executed by using the conversion tables **106**, but also information and programs for executing other types of processing, such as startup processing and security processing. Details of content codes **105** are described below.

An overview of the configurations of the host **140** and the drive **120** and processing performed by thereof is discussed below with reference to FIG. **1**. Content stored in the information recording medium **100** is played back by transferring data concerning the content to the host **140** via the drive **120**.

In the host **140**, the playback (player) application **150** and a secure VM **160** are set. The playback application **150** is a content playback processor, and executes processing, such as authentication with the drive **120**, content decryption, and decoding.

The secure VM **160** executes processing by using the content codes **105** including the conversion tables **106** and the security check codes **107**. The secure VM **160** selects the security check code **107** corresponding to the associated player from the security check codes **105** to perform security check processing, and also performs replacement processing for replacing part of the data forming the content by using the conversion tables **106**. The secure VM **160** is set as a virtual machine in the host **140**. The virtual machine (VM), which is a virtual computer that directly analyzes an intermediate language and executes it, reads instruction code information of a platform-independent intermediate language, from the information recording medium **100**, and analyzes the instruction code information to execute it.

The secure VM **160**, which serves as a data processor that obtains the content codes **105** containing a program or information for using the encrypted content **101** recorded on the information recording medium **100** and that executes the content codes **105** to perform data processing.

Communication between the playback application **150** and the secure VM **160**, such as sending and receiving information and processing requests, is performed by a sequence of interrupt (INTRP) from the playback application **150** to the secure VM **160** and response (call) from the secure VM **160** to the playback application **150**.

Major processing executed by the host **140** is as follows. Before the use of content, mutual authentication processing between the drive **120** and the host **140** is conducted. After verifying the integrity of the drive **120** and the host **140** as a result of mutual authentication processing, the encrypted content **101** is transferred from the drive **120** to the host **140**, and the host **140** decrypts the encrypted content **101** and also executes data conversion processing by using the conversion tables **106** to play back the content.

A data processor **121** of the drive **120** conducts authentication with the host **140** to use the content, and reads the data from the information recording medium **100** and transfers it to the host **140**.

The playback (player) application **150** of the host **140**, which is a data playback (or recording) application executed by an information processing apparatus, for example, a PC, executes processing by using hardware of the information processing apparatus in accordance with a predetermined data processing sequence.

The host **140** includes a data processor **151** that executes mutual authentication with the drive **120** and data transfer control, a decrypt processor **153** that decrypts the encrypted content **101**, a data conversion processor **154** that performs data conversion processing on the basis of the data registered in the conversion tables **106**, and a decode processor **155** that performs decoding (for example, MPEG decoding).

The decrypt processor **153** generates a key used for decrypting the encrypted content **101** by using various information stored in a memory **156** and the data read from the information recording medium **100** to decrypt the encrypted content **101**. The data conversion processor **154** executes replacement (overwrite) processing for replacing (overwriting) part of the data forming the content by using the conversion data registered in the conversion tables **106** in accordance with the data conversion processing program obtained from the information recording medium **100**. The decode processor **155** executes decoding (for example, MPEG decoding) processing.

In the memory **156**, a device key (Kd) and key information used for mutual authentication and decryption are stored. The device key (Kd) is a key used for the above-described MKB processing. The MKB **102** is a key information block that makes it possible to obtain a media key (Km), which is a key necessary for decrypting content, only by processing (de-

cryption) by using the device key (Kd) stored in information processing apparatuses having valid licenses. To decrypt the encrypted content **101**, the playback application **150** (information processing apparatus) executes the MKB processing by using the device key (Kd) stored in the memory **156**. Details of content decryption processing are given below.

2. Content Management Units (CPS units)

As stated above, to control the use of content based on different content units, content stored in an information recording medium is encrypted and stored by assigning different keys to content units. That is, the content is segmented into content management units (CPS units), and the CPS units are individually encrypted and usage control is performed for the individual CPS units.

To use content, it is necessary to first obtain a CPS unit key (also referred to as a "title key") assigned to each unit, and then, by using the CPS unit key and other necessary keys and key generation information, data processing based on a predetermined decrypt processing sequence is executed to play back content. An example of the setting of content management units (CPS units) is discussed below with reference to FIG. **2**.

Content has, as shown in FIG. **2**, a hierarchical structure including (A) an index layer **210**, (B) a movie object layer **220**, (C) a playlist layer **230**, and (D) a clip layer **240**. When designating an index, such as a title, accessed by a playback application, a playback program associated with the title is designated, and a playlist defining, for example, a content playback order, is selected according to program information concerning the designated playback program.

A playlist includes one or more play items as information concerning zones to be played back. According to clip information as a playback zone defined by the play items, AV streams or commands as real content data are selectively read to play back the AV streams or to execute the commands. There are a plurality of playlists or a plurality of play items, and playlist IDs or play item IDs are associated with the playlists or play items as identification information.

FIG. **2** shows two CPS units, which form part of content stored in an information recording medium. Each of a CPS unit-**1 271** and a CPS unit-**2 272** includes a title as an index, a movie object as a playback program file, a playlist, and an AV stream file as real content data.

The content management unit (CPS unit)-l **271** includes a title-**1 211** and a title-**2 212**, playback programs **221**, **222**, and **223**, playlists **231** and **232**, and clips **241** and **242**. At least AV stream data files **261** and **262**, which serve as real content data, contained in the two clips **241** and **242**, respectively, are data to be encrypted, and are basically encrypted with a title key (Kt**1**) (also referred to as the "CPS unit key"), which is a cryptographic key associated with the content management unit (CPS unit)-l **271**.

The content management unit (CPS unit)-**2 272** includes an application-l **213** as an index, a playback program **224**, a playlist **233**, and a clip **243**. An AV stream data file **263**, which serve as real content data, contained in the clip **243**, is encrypted with a title key (Kt**2**), which is a cryptographic key associated with the content management unit (CPS unit)-**2 272**.

To play back an application file or content associated with the content management unit (CPS unit)-**1 271**, the user needs to obtain the title key (Kt**1**), which is the cryptographic key set for the content management unit (CPS unit)-**1 271**, to perform decrypt processing. To play back an application file or content associated with the content management unit (CPS unit)-**2 272**, the user needs to obtain the title key (Kt**2**), which is the cryptographic key set for the content management unit (CPS unit)-**2 272**, to perform decrypt processing.

An example of the setting of CPS units and the association between the CPS units and the title keys is shown in FIG. **3**. More specifically, FIG. **3** illustrates the association between elements for setting the CPS units, which are the management units for using encrypted content stored in an information recording medium, and the title keys (CPS unit keys) used for the CPS units. It is possible to store beforehand CPS units and title keys to be used for data generated or obtained by users afterwards. For example, a data portion **281** is an entry for data generated or obtained by users afterwards.

Various elements can be used for setting CPS units, for example, content titles, applications, data groups, etc. In a CPS unit management table, CPS unit IDs, which serve as identifiers associated with the CPS units, are set.

In FIG. **3**, for example, title **1** belongs to CPS unit **1**, and to decrypt encrypted content belonging to CPS unit **1**, it is necessary to generate title key (Kt) and to decrypt the content by using the generated title key (Kt**1**).

As discussed above, to individually control for content units, content management units (CPS units) forming the content stored in the information recording medium **100** are encrypted with different keys. To individually control the usage of each CPS unit, usage rules (URs) are set for each CPS unit. As stated above, the usage rules include copy/playback control information (CCI) for content, such as copy restriction information and playback restriction information concerning encrypted content included in each CPS unit.

To generate title keys, data processing using various items of information stored in an information recording medium is necessary. Specific examples of such processing are described in detail below.

3. Overview of Structure of Content Data Including Modified Data and Data Conversion Processing

An overview of the structure of content data including modified data and data conversion processing is described below. As stated above, in the encrypted content **101** stored in the information recording medium **100**, part of the data forming the encrypted content **101** is replaced by data different from correct data and is set as broken data. With this arrangement, even if the encrypted content **101** including the broken data is decrypted, it is difficult to play back content correctly. To play back content correctly, it is necessary to replace (overwrite) the broken data by conversion data registered in the conversion tables **106**.

An overview of the structure of content stored in the information recording medium **100** and playback processing is discussed below with reference to FIG. **4**. In the information recording medium **100**, AV content, for example, a movie, is stored. The content is encrypted, and can be played back after being decrypted with a cryptographic key that can be obtained only by a playback apparatus with an authorized license. Specific content playback processing is discussed below. In the content stored in the information recording medium **100**, part of the data is replaced by modified data.

FIG. **4** illustrates the structure of content **291** stored in the information recording medium **100**. The content **291** includes correct content **292** without being modified and broken data **293**, which is modified. The broken data **293** is generated by destroying the original correct content. Accordingly, content playback cannot be correctly performed by using the content **291** including the broken data **293**.

To correctly perform content playback, it is necessary to generate playback content **296** by replacing the broken data **293** contained in the content **291** by correct content data. To generate the playback content **296**, conversion data, which

serves as correct content data to be substituted for the broken data **293** in the corresponding broken data areas, are obtained from conversion entries **295** registered in the conversion tables (fix-up tables) **106** contained in the content codes **105** (see FIG. **1**), and the broken data **293** are replaced by the obtained conversion data. Then, the playback content **296** is played back.

When generating the playback content **296**, in addition to the replacement of the broken data **293** by conversion data **297**, which is correct content data, part of the content **291** is replaced by identifier-setting conversion data **298** for analyzing bits forming identification information (for example, a player ID), which is used for identifying a content player or a content playback application. With this replacement, if illegally copied content is distributed, the distribution source of unauthorized content can be specified by analyzing the identifier-setting conversion data **298** in the distributed content.

The conversion entries **295** are stored in the conversion tables **106**, and they may also be dispersedly recorded in specific packets in the data forming the content. That is, multiple recording of the conversion entries **295** in the conversion tables **106** and in the encrypted content **101** shown in FIG. **1** is implemented. An information processing apparatus to play back content performs replacement processing by using either the conversion data stored in the conversion tables **106** or the conversion entries **295** dispersedly recorded in the content.

4. Content Playback Processing

Content playback processing executed by a host **345** is described below with reference to FIG. **5**. In FIG. **5**, there are shown an information recording medium **330** storing encrypted content therein, a drive **340** that sets the information recording medium **330** and reads data therefrom, and the host **345** connected to the drive **340** so that they can communicate with each other to execute a playback (player) application which obtains content stored in the information recording medium **330** via the drive **340** and plays back content.

In the host **345** shown in FIG. **5**, a playback application block **350** that executes content decryption, decoding, and data conversion processing and a secure VM block **360** having a secure VM **361** are separated. The secure VM **361** calculates parameters used for conducting security check based on security check codes **335** contained in content codes **334** recorded on the information recording medium **330** and used for conversion processing based on conversion tables (fix-up tables) **336**.

The information recording medium **330** includes an MKB **331**, a title key file **332**, encrypted content **333**, and the content codes **334** as recorded data. In the encrypted content **333**, as discussed with reference to FIG. **4**, part of data is substituted with broken data, and it is necessary to replace the broken data by data obtained from the conversion tables **336**.

The content codes **334** include the security check codes **335** and the conversion tables **336**. The security check codes **335** include a program for verifying the integrity of a player (playback apparatus) to perform content playback. The conversion tables **336** store conversion data corresponding to data to be replaced in predetermined areas of the content. The host **345** stores a device key **351** used for MKB processing.

The playback processing sequence performed by the host **345** for obtaining the content stored in the information recording medium **330** via the drive **345** and playing back the content is as follows. Before reading the content from the information recording medium **330**, in step S**101**, the host **345** and the drive **340** conduct mutual authentication to check whether they are an authorized application and apparatus. The mutual authentication can be conducted in various manners. After the mutual authentication has been successfully conducted, the drive **340** and the host **345** share a common session key (Ks), which is a common private key.

Then, in step S**102**, the playback application **350** of the host **345** obtains the MKB **331** recorded on the information recording medium **330** via the drive **340** and executes processing on the MKB **331** by using the device key **351** stored in a memory to obtain a media key (Km).

The MKB **331** is a cryptographic key block generated based on a tree-structure key distribution system, which is known as one mode of the broadcast encryption method. The MKB **331** is a key information block that makes it possible to obtain the media key (Km), which is a key necessary for decrypting content, only by processing (decryption) by using a device key (Kd) stored in information processing apparatuses having valid licenses.

Then, in step S**103**, the playback application **350** decrypts the title key file **332** read from the information recording medium **330** by using the media key (Km) obtained in the MKB processing executed in step S**102** to obtain a title key (Kt). The title key file **332** stored in the information recording medium **330** is a file including data encrypted with the media key (Km), and by decrypting the title key file **332** by using the media key (Km), the title key (Kt) used for decrypting the content can be obtained. The decryption processing in step S**103** can be executed by using, for example, Advanced Encryption Standard (AES) encryption algorithms.

Then, the playback application **350** reads the encrypted content **333** stored in the information recording medium **330** via the drive **340** and stores the read encrypted content **333** in a track buffer **352**. Then, in step S**104**, the playback application **350** decrypts the encrypted content **333** stored in the track buffer **352** by using the title key (Kt) to obtain decrypted content.

The decrypted content is stored in a plaintext transport stream (plain TS) buffer **353**. The decrypted content stored in the plain TS buffer **353** contains the above-described broken data, and thus, predetermined data conversion (data replacement by overwriting) is necessary.

In step S**105**, the secure VM **361** generates parameters necessary for data conversion from the content codes **334**. Then, in step S**106**, table-reconstruction/data-conversion processing is executed under the control of a real time event handler **356**. By the control of the real time event handler **356**, the playback application **350** outputs a parameter calculation request to the secure VM **361** as an interrupt (INTRP) in response to the switching of content segments. Upon sequentially receiving parameters from the secure VM **361**, the playback application **350** decrypts or computes conversion table blocks to obtain plain conversion table blocks and extracts conversion entries contained in the obtained conversion table blocks.

In the conversion entries, (a) conversion data, (b) identifier-setting conversion data, and recording-position specifying information concerning the recording positions of the conversion data in the content are recorded. In step S**106**, the playback application **350** executes data conversion processing for writing the conversion data and the identifier-setting conversion data into the specified positions in real time, simultaneously with the content playback processing or the output processing.

The secure VM **361** generates different parameters for the individual content segments in accordance with the content codes **334** and outputs the generated parameters. If parameters SP**1**, SP**2**, SP**3**, . . . , are exclusive OR (XOR) computation parameters with the conversion entries corresponding to

the content segments, the following exclusive computation operations are executed as the table reconstruction processing in step S106:

[Conversion entry 1] (XOR) [SP1]

[Conversion entry 2] (XOR) [SP2]

[Conversion entry 3] (XOR) [SP3], and so on.

Then, the conversion entries contained in the conversion table block data are obtained. In the above computation operations, [A] (XOR) [B] means exclusive OR (XOR) computation between A and B.

In this manner, the conversion entries included in the content 333 recorded on the information recording medium 330 have been subjected to exclusive OR (XOR) computation with the parameters SP1, SP2, SP3, and so on. Those parameters SP1, SP2, SP3, and so on, are obtained and output by the secure VM 361.

In the table-reconstruction/data-conversion processing in step S106, conversion data is obtained from the conversion entries reconstructed by the computation or encryption using the parameters SP1, SP2, SP3, and so on, and the broken data included in the content is replaced by the obtained conversion data, which is correct content data. Additionally, part of the data is also replaced by the identifier-setting conversion data. Then, the data stored in the plain TS buffer 353 is changed to converted data. An overview of this data conversion processing is discussed below with reference to FIG. 6.

The encrypted content 333 stored in the information recording medium 330 is temporarily stored in the track buffer 352 of the host 350. The encrypted content 333 stored in the track buffer 352 corresponds to track buffer storage data 401 indicated by (1) in FIG. 6. The track buffer storage data 401 is decrypted into decrypted data 402, indicated by (2) in FIG. 6, by the host 350, and is stored in the plain TS buffer 353.

The decrypted data 402 contains broken data 403, which is not correct content data. The data conversion processor of the host 350 thus replaces the broken data 403 by conversion data 404, which is the correct content data. This replacement processing is executed by rewriting (overwriting) part of the data 402 stored in the plain TS buffer 353.

In addition to the replacement processing for replacing the broken data 403 by the conversion data 404, which is the correct content data, the host 350 replaces, as shown in FIG. 6, part of the decrypted data 402 by identifier-setting conversion data 405.

As stated above, the identifiers are data that can analyze bits forming the identification information for identifying a content playback apparatus or a content playback application. More specifically, the identifiers are identification information (player ID) concerning an information processing apparatus, which is a player executing a host application, or identification marks generated based on the player ID. The identifier-setting conversion data 405 is generated by slightly modifying the bits forming the correct content data to such a degree as not to influence playback of the content.

A plurality of the identifier-setting conversion data 405 are set in the content, and by collectively analyzing the identifier-setting conversion data 405, the player ID, for example, can be identified. The identifier-setting conversion data 405 is data that can identify bits forming identification marks by MPEG bit stream analysis.

In the conversion tables 336 stored in the information recording medium 330, many conversion data 404 and identifier-setting conversion data 405 shown in FIG. 6 are registered, and also, information concerning the recording positions of the conversion data 404 and the identifier-setting conversion data 405 is registered. By executing the data conversion processing based on the information stored in the conversion tables 336, the data stored in the plain TS buffer 353 is replaced by converted data 406 indicated by (3) in FIG. 6.

Then, the converted TSs are output to an external source via a network, and are played back by an external playback apparatus. Alternatively, in step S107, the TSs are converted into elementary streams (ESs) by a demultiplexer, and in step S108, the ESs are decoded. The decoded ESs are played back by a display speaker.

5. Processing Using Security Check Codes

Before starting the above-described content playback processing, the-secure VM 361 conducts security check by using the security check codes 335 contained in the content codes 334. If necessary, the secure VM 361 continues security check while content is being played back.

The secure VM 361 verifies the integrity of a player (playback apparatus) in accordance with the security check codes 335 contained in the content codes 334 under the control of an event handler 354. As stated above, the conversion tables 336 and the security check codes 335 contain various types of codes to allow the execution of processing in accordance with the types of players and player port information.

The secure VM 361 obtains information concerning the player (player information 355), such as a player certificate stored in the storage unit of the player and information concerning ports of the player, and selects the security check code corresponding to the player from the security check codes 335 to conduct security check. That is, the secure VM 361 obtains, as the player information, at least one of the identification information and the attribute information corresponding to an information processing apparatus or a content usage application, and selects the security check code corresponding to the player information to execute security check based on the selected code.

As discussed above, when using content stored in an information recording medium, the secure VM 361 conducts security check. After the integrity of the player is verified as a result of the security check and after the denial of unauthorized output of content is ensured based on, for example, the player information 355, the content is played back. If the configuration of the player is changed, for example, if a new device is connected to a port of the player, security check may be conducted in accordance with the changed configuration of the player.

Different types of security check may be necessary in accordance with the configurations of playback apparatuses or the types of playback applications. Accordingly, the security check codes 335 are recorded in the content codes 334 as a set of codes corresponding to various players and applications.

The structure of content codes recorded in an information recording medium is discussed below with reference to FIGS. 7 through 9. FIG. 7 illustrates the directory structure of the overall data stored in an information recording medium. The storage data in an information recording medium is largely divided into two data: one is a BDMV directory including content-related data, such as content management data, CPS unit key, content usage control information (CCI), and content, and the other is a BDSVM directory including content codes, such as security check codes and conversion tables.

Details of the BDMV directory and the BDSVM directory are discussed below with reference to FIGS. 8 and 9, respectively. When storing content having a hierarchical structure discussed with reference to FIG. 2 in an information recording medium, various data and programs, such as content codes, are recorded as individual files in the information

recording medium in accordance with, for example, the directory structure shown in FIG. **8**. The files stored in the information recording medium are, for example, as follows:

(A) index.bdmv file corresponding to the index layer **210** shown in FIG. **2**;

(B) Movieobject.bdmv file corresponding to the movie object layer **220** shown in FIG. **2**;

(C) files under a PLAYLIST directory corresponding to the playlist layer **230** shown in FIG. **2**;

(D) files under a CLIPINF directory and files under a STREAM directory corresponding to the clip layer **240** shown in FIG. **2**, the files under the CLIPINF directory and the STREAM directory shown in FIG. **8** being associated with the clips and the streams shown in FIG. **2** according to the same file numbers; and

(E) other files, such as an AUXDATA File storing sound data and font data, a META file storing metadata, and a BDJO file storing BD-J objects.

In content stored in an information recording medium, as stated above, part of the data is replaced by broken data, which is different from correct content data. To correctly play back the content, it is necessary to replace the broken data by data (converted data) registered in a conversion table. In this replacement processing, the broken data is converted into the data registered in the conversion tables (fix-up table) by using the content codes stored in the information recording medium.

The content codes including the conversion tables and the security check codes are also stored in the information recording medium as individual files. The directory structure including the content codes is shown in FIG. **9**. The directory structure shown in FIG. **9** is the structure including the content codes created for AV content having the directory structure shown in FIG. **8**.

As stated above, the content codes include security check codes and conversion tables. The content codes in an information recording medium are stored, as shown in FIG. **9**, in a plurality of individual files nnnnn.svm set in the BDSVM directory. Backup data is also set in a BACKUP directory as copy data.

The files storing the content codes are classified into the following categories:

(a) content codes common to all items of content and all players (playback apparatuses or playback applications);

(b) content codes unique to content;

(c) content codes unique to players (playback apparatuses or playback applications); and

(d) content codes unique to content and players (playback apparatuses or playback applications).

By classifying the content codes into the categories (a) through (d), the content codes can be set as independent data files, and thus, the data files can be reused. That is, some content code files can be reused in common for different items of content and different players (playback apparatuses or playback applications). The reuse of content codes is discussed below with reference to FIG. **10**.

In FIG. **10**, content code files **601** through **604** are files kept by content code production or providing entities. The content code files **601** through **604** are a content/player-common content code file **00000**.svm **601**, player-unique content code files **00001**.svm and **00002**.svm **602**, a content-unique content code file **00003**.svm **603**, and a content/player-unique content code file **00004**.svm **604**, respectively.

After attaching digital signatures of the content code production or providing entities to the content code files **601** through **604**, they are kept in the content code production entities or providing entities.

When producing an information recording medium storing new content, each entity can reuse the content code files **601** through **604** that have already been used for another content. If, for example, a content-unique content code file and a player-unique content code file are set as the same file, and in this case, if content is changed, the content-unique content code file has to be changed. This means that the content code file common to the content-unique content code file and the player-unique content code file is changed even if it is not necessary to change the player-unique content code. Accordingly, since a digital signature is attached to the content code file, it should also be changed, which hampers the reuse of the payer-unique content code file. In contrast, if the content code files are individually created according to the categories, as shown in FIG. **10**, the player-unique content code files, for example, can be reused when another disk is manufactured.

To prevent tampering of the content codes, each content code file is provided to a management center (KIC), and is provided with a digital signature and is stored in an information recording medium **610**. The management center attaches digital signatures and unique IDs to content codes to be recorded in the information recording medium **610**. Content codes **620** recorded on the information recording medium **610** include, as shown in FIG. **10**, security check codes **621** and conversion tables (fix-up tables) **622**. In the specific directory structure, content codes generated by individual entities are set, as indicated by a directory structure **630** shown in FIG. **10**.

In this manner, content codes can be reused for various items of content, and content codes that can be reused and content codes that should be changed are suitably combined, and are recorded in an information recording medium.

The content code files may be set, as shown in FIG. **9**, according to the following categories:

content code file **00000**.svm: code for determining player information;

content code file **00001**.svm and **00002**.svm: codes selected according to player information (e.g., **00001**.svm is code for player A and **00002**.svm is code for player B); and

content code file **00003**.svm: code that does not depend on player information (e.g., default code indicated in **00003**.svm is used for players to be sold after the content is released).

As discussed above, different content codes classified into various categories are stored in an information recording medium, and a player (playback apparatus) to conduct security check using content codes selects the security check code associated with the player.

The secure VM **361** shown in FIG. **5** selects the security check code corresponding to the player associated with the secure VM **361** to conduct security check. In this case, the secure VM **361** obtains the player information **355** to conduct security check by using the security check code.

The player information **355** includes a player certificate stored in the memory of the player and information concerning the ports of the player. The player information **355** include various types of information, such as information that can be directly obtained by the secure VM **361** from the memory, information that can be obtained via a playback application, and information that can be obtained via an operating system (OS).

For example, information, such as a player certificate, can be obtained via a playback application. However, information concerning players, such as the number of ports provided for a player, or information concerning a device connected to a port, is not necessarily obtained directly by a playback application, and can be obtained only via another program, such as

an OS. The secure VM **361** obtains information directly or via a playback application or another program to conduct security check.

One type of security check executed by the secure VM **361** is to check whether a playback apparatus has an authorized player certificate. The player certificate certifies the usage right of content and is issued by a management entity that manages the content.

An example of the configuration of the player certificate is shown in FIG. **11**. The player certificate includes, as shown in FIG. **11**, data indicating a player certificate size, a certificate version, a player manufacturer ID, a serial number, a signature time and date, a device (player) attribute information, a player public key, and a digital signature. Data other than the above-described data, such as that concerning a player model name and a player model version, may be included.

The secure VM **361** verifies the player certificate in accordance with the security check codes **335** read from the information recording medium **330**, and after the integrity of the player certificate is verified, the secure VM **361** can obtain information necessary for further security check from this certificate. More specifically, the security VM **361** first checks the integrity of the signature attached to the player certificate by using, for example, a public key of the management center, which is the entity that has attached the signature to the player certificate. The public key of the management center may be stored beforehand in the memory of the playback apparatus or may be obtained from the information recording medium **330** directly or via a network.

If the integrity of the player certificate is not verified, processing to perform content playback accompanying data conversion is discontinued. If the integrity of the player certificate is verified, security check is conducted for the player. Basic player information, such as a manufacturer, can be obtained from the player certificate.

That is, the secure VM **361** checks the integrity of the player certificate, and under the condition that the integrity of the player certificate is verified, the secure VM **361** obtains identification information or attribute information corresponding to the information processing apparatus or the content usage application, such as the manufacturer, type, version, and serial number of the apparatus or the application, from the information recorded on the player certificate. Based on the obtained information, the secure VM **361** selects the security check code corresponding to the obtained information to conduct security check based on the selected code. The player information, such as information necessary for security check, is obtained by the playback application **350** or the secure VM **361**.

Player information (player configuration information) other than a player certificate is described below with reference to FIG. **12**. Player information used for conducting security check includes, as shown in FIG. **12**, the following information:

- port attributes: one-way, two-way, analog, digital, video, audio, recordable/not-recordable;
- port statuses: format of data output via a port (composite video, component video, digital audio, and digital video) and protocol information (VGA/HDCP/DTCP); and
- player information: RAM size, system clock, and data conversion (media transform) processing method for the virtual machine (VM).

When checking for the possibility of the unauthorized use of content, the secure VM **361** may need to check whether unauthorized output or copying of content using a specific port has been conducted. In this case, the secure VM **361** refers to port attributes and port statuses obtained as the player information **355**, and selects a suitable security check code corresponding to the player information **355** to check the unauthorized use of content. The selection and application of a suitable security check code by using the player information **355** implements efficient and effective security check using content codes. That is, a certain execution time is necessary for executing content codes. Accordingly, by executing a necessary security check code selected from a plurality of security check codes stored in a disk, security check can be conducted speedily without decreasing the security level. The execution speed of security check codes can be ensured as follows. A player is first specified, and a security check code is selected based on the performance of the specified player so that a player having higher performance is allowed to execute more security check codes than a player having lower performance. With this arrangement, the user does not have to wait while the security check codes are being executed. More specifically, the time used for executing security check codes is first determined, and then, security check codes are generated and selected so that the execution of the security check codes can be finished within the given time.

The player information **355** is obtained by the playback application **350** or the secure VM **361** via various routes. The necessary player information **355** is obtained, and the secure VM **361** executes the security check codes **355** by using the player information **355** to conduct security check.

Processing for obtaining player information is discussed below with reference to FIG. **13**. FIG. **13** illustrates layers forming a playback apparatus, for example, a PC or a playback-only device (consumer electronics (CE) apparatus), which executes content playback. In FIG. **13**, there are shown a process execution application, such as a playback application **701** that executes content playback, which is the highest layer, an OS **711** under the playback application **701**, and a hardware/device driver **721** including various hardware and device drivers, under the OS **711**.

A secure VM **702**, which is generated by the playback application **701** as a virtual machine, conducts security check using security check codes. The secure VM **702** conducts security check by obtaining various player information. For example, the player certificate shown in FIG. **11** is stored in the memory of the playback apparatus, and is contained in player information **703** that can be directly obtained by the playback application **701**. Other information, such as device/application-dependent information, e.g., the number of ports of a CE device, is also contained in the player information **703**.

In contrast, information concerning various devices connected to the ports of the playback apparatus is not contained in the player information **703** that can be directly obtained by the playback application **701**. Such information includes device information concerning the hardware/device-driver **721**, and is contained in player information **713** that can be obtained by the OS **711**.

The playback application **701** obtains the player information **713** from the OS **711** and provides it to the secure VM **702** that executes security check.

The player information **713** that can be obtained by the playback application **701** from the OS **711** includes information that may be changed by, for example, the extension of the system or the connection of an external device. Such information can be obtained from a device manager **712** of the OS **711** while the playback application **701** is being executed, and includes information concerning, for example, platforms and ports.

It is now assumed that, when the playback application **701** is started, it checks the condition of the apparatus, and then performs content playback in accordance with the configuration of the apparatus. In this case, the playback application **701** obtains information concerning the condition of the apparatus and provides the information to the secure VM **702**. If the playback application **701** has the function of obtaining the latest information concerning, for example, a new device connected to the apparatus after the playback application **701** is started, it provides the updated information to the secure VM **702** if necessary.

There is some device information, such as that indicated by C**2** in FIG. **13**, which cannot be obtained by the playback application **701** from the OS **711**. For example,. it is difficult for a general playback application to obtain the register value stored in a device driver unless it executes special processing with the aid of the device driver and the OS **711**.

Basically, the playback application **701** does not obtain such information, and conducts security check by using only available player information, such as a player certificate and apparatus information.

A description is now given, with reference to the flowchart in FIG. **14**, of a content playback sequence including security check processing using security check codes included in content codes and conversion processing using conversion tables.

In step S**301**, the secure VM obtains content codes from an information recording medium and determines whether a request to obtain player information is included in the content codes. For example, the secure VM reads the content code file **00000**.svm that stores a code for determining player information in the directory structure shown in FIG. **9**.

If it is determined in step S**301** that the request to obtain player information is not included in the content codes, the process proceeds to step S**306**. If the request to obtain player information is included in the content codes, the process proceeds to step S**302** to obtain player information necessary for security check. The player information includes, for example, a player certificate, such as that shown in FIG. **11**, and port attributes, port statuses, and other player information, such as those shown in FIG. **12**. As discussed with reference to FIG. **13**, an apparatus having expansion functions, such as a PC, obtains player information from the device management function of an OS, to acquire port attributes. A CE device obtains fixed values as player information since port attributes for the CE device are fixed. In this manner, necessary player information is different depending on the type of apparatus, and also, player information is obtained in different manners.

In step S**303**, it is determined whether a security check code corresponding to the obtained player information is included in the content codes. If it is determined in step S**303** that the security check code is included, the process proceeds to step S**304** to conduct security check corresponding to the security check code. Security check is conducted by using, for example, the content code file **00001**.svm or **00002**.svm, shown in FIG. **9**, which is selected according to the player information.

It is then determined in step S**305** as a result of security check whether there is any problem. If a problem is found, processing to perform content playback is terminated. If there is no problem as a result of security check, the process proceeds to step S**306** to determine whether data conversion processing using a conversion table contained in the content codes is necessary. If data conversion processing is not necessary, the process proceeds to step S**308** to perform content playback.

If it is determined in step S**306** that data conversion processing is necessary, the process proceeds to step S**307** to obtain a conversion table and to decrypt it to perform content conversion processing by using information recorded in the conversion table. This conversion processing corresponds to the processing discussed with reference to FIG. **6**. Then, in step S**307**, content is played back. It should be noted that data conversion processing in step S**307** and content playback processing in step S**308** are simultaneously performed.

6. Configuration of Information Processing Apparatus

A description is now given, with reference to FIG. **15**, of the hardware configuration of an information processing apparatus (host) **800** that executes data processing using the above-described playback (player) application and secure VM. The information processing apparatus **800** includes a central processing unit (CPU) **809**, a ROM **808**, which serves as a storage area storing programs and parameters therein, a memory **810**, a digital input/output interface (I/F) **802** that inputs and outputs digital signals, an analog input/output I/F **804** that inputs and outputs analog signals and includes analog-to-digital (A/D) and digital-to-analog (D/A) converters **805**, an MPEG codec **803** that encodes and decodes MPEG data, a transport-stream (TS)/program-stream (PS) processor **806** that performs TS/PS processing, an encryption unit **807** that performs mutual authentication and various types of encryption processing, such as decryption of encrypted content, a recording medium **812**, such as a hard disk, and a drive **811** that drives the recording medium **812** and inputs and outputs data recording/playback signals. The above-described elements are connected to a bus **801**. The CPU **809** performs data processing in accordance with various programs including various types of processing that accompany processing performed by an OS or a content playback/recording application program, mutual authentication processing, and content playback, for example, security check processing based on security check codes and data conversion processing using conversion tables.

The information processing apparatus **800** is connected to the drive **811** via a connection bus, such as an advanced technology attachment packet interface (ATAPI)-BUS bus. Conversion tables and content are input or output via the digital input/output I/F **802**. The encryption processor **807** performs encryption processing and decryption processing by using, for example, AES algorithms.

The program for executing content playback/recording processing is stored in, for example, the ROM **808**, and the memory **810** is used as a work area or for storing parameters and data while the program is being executed.

In the ROM **808** or the recording medium **812**, the above-described player certificate, a public key of a management center which is used for verifying the signature of the player certificate, a host private key and a host public key for conducting authentication with the drive **811**, and a revocation list of revoked public key certificates are stored.

To play back content or to output content to an external source, by using a data conversion processing program obtained from the information recording medium **812**, decryption of encrypted content, reconstruction of conversion tables, and overwriting of conversion data based on the data stored in the conversion tables are performed in accordance with the above-described processing sequence.

7. Information Recording Medium Manufacturing Apparatus and Information Recording Medium

An information recording medium and an information recording medium manufacturing apparatus, and more specifically, an information recording medium used in the above-

described content playback processing and a manufacturing apparatus and method for such an information recording medium, are discussed below.

The information recording medium manufacturing apparatus is an apparatus for manufacturing, for example, the information recording medium **100** storing the recorded data described with reference to FIG. **1**. The information recording medium manufacturing apparatus includes, as shown in FIG. **16**, a content file generator **901** that generates content files storing content data to be recorded on an information recording medium **910**, a content code file generator **902** that generates content code files storing content codes including a security check processing program to be executed for using content, and a recorder **903** that records the content files generated by the content file generator **901** and the content code files generated by the content code file generator **902** on the information recording medium **910**.

The content code file generator **902** generates, as discussed with reference to FIG. **9**, a plurality of content code files in accordance with various types of information processing apparatuses and content usage applications.

The content code file generator **902** also generates content code files storing common content codes that do not depend on the type of information processing apparatus or content usage application, content code files storing content codes depending on the type of information processing apparatus or content usage application, and content code files storing codes for determining the types of information processing apparatuses or content usage applications. The recorder **903** records those various content code files on the information recording medium **910**.

When generating a plurality of content code files storing content codes classified into different categories, the content code file generator **902** generates files, all of which are provided with digital signatures, or files, only one of which is provided with a digital signature. The digital signatures attached to the files generated by the content code file generator **902** may be only digital signatures of a management center (KIC) or digital signatures of a management center and digital signatures of content code production or providing entities.

On the information recording medium **910** generated by the information recording medium manufacturing apparatus, various data discussed above are recorded. More specifically, at least content files storing content data and content code files storing content codes including a security check processing program to be executed for using content are recorded. The content code files include a plurality of content code files associated with the types of information processing apparatuses or content usage applications.

Content code files recorded on the information recording medium **910** may be classified, as discussed with reference to FIG. **9**, into code files storing common content codes that do not depend on the type of information processing apparatus or content usage application, code files storing content codes depending on the type of information processing apparatus or content usage application, and code files storing codes for determining the types of information processing apparatuses or content usage applications.

The content code files to be recorded on an information recording medium include a plurality of content code files storing content codes classified into different categories. Among the plurality of recorded files, only one file may be provided with a digital signature, or all the files may be provided with digital signatures. Digital signatures of content code production or providing entities may be attached to the content code files.

A series of processing operations described in the specification may be executed by hardware, software, or a combination thereof. If software is used, a program storing processing sequences therein may be installed into a memory of a computer built in dedicated hardware or into a general-purpose computer that can execute various types of processing, and is then executed.

The program may be stored beforehand in a hard disk or a ROM, which serves as a recording medium. Alternatively, the program may be temporarily or permanently stored (recorded) in a removable recording medium, such as a flexible disk, a compact disc read only memory (CD-ROM), a magneto-optical disk (MO disk), a digital versatile disc (DVD), a magnetic disk, or a semiconductor memory. Such a removable recording medium can be provided as so-called "package software".

As stated above, the program may be installed into a computer from the above-described removable recording medium. Alternatively, the program may be transferred from a download site to a computer wirelessly or by wired means via a network, such as a local area network (LAN) or the Internet. In this case, the computer receives the program and installs it into a built-in recording medium, such as a hard disk.

Various processing operations described in the specification may be executed in chronological order discussed in the specification. Alternatively, they may be executed in parallel or individually in accordance with the performance of an apparatus executing the processing or according to the necessity. In this specification, the system is a logical set of a plurality of devices, and it is not essential that the devices be in the same housing.

It should be understood by those skilled in the art that various modifications, combinations, sub-combinations and alterations may occur depending on design requirements and other factors insofar as they are within the scope of the appended claims or the equivalents thereof.

What is claimed is:

1. An information processing apparatus that performs playback processing for content recorded on an information recording medium, comprising:

a data processor that, when using the content recorded on the information recording medium comprising the content and content codes that comprise security check codes to conduct a security check for the information processing apparatus:

in response to a determination by the data processor of the information processing apparatus that the content codes include a request to obtain, from memory of the information processing apparatus, identification information related to a player certificate or a port status of the information processing apparatus, obtains from the memory of the information processing apparatus the identification information comprising certificate information of the information processing apparatus or configuration information comprising the port status of the information processing apparatus; and

determines, based on the request, whether the security check codes classified into a plurality of data files comprise a security check code corresponding to the identification information to conduct a security check on the basis of the security check code.

2. The information processing apparatus according to claim **1**, wherein the data processor obtains the player certificate stored in a storage unit of the information processing apparatus and verifies an integrity of the player certificate,

and if the integrity of the player certificate is verified, the data processor obtains the identification information from information recorded on the player certificate.

3. The information processing apparatus according to claim 1, wherein the configuration information further comprises at least one of port attribute information, memory area size information, and data processing mode information concerning the information processing apparatus, and wherein the information processing apparatus conducts security check corresponding to the obtained configuration information.

4. The information processing apparatus according to claim 1, wherein, when obtaining the configuration information, the data processor obtains device information managed by an operating system and conducts security check corresponding to the obtained device information.

5. The information processing apparatus according to claim 1, wherein, when the configuration information is updated, the data processor obtains the updated configuration information and conducts security check corresponding to the updated configuration information.

6. The information processing apparatus according to claim 1, wherein the data processor performs data processing which accompanies content data replacement processing using a conversion table included in the content codes, the data processing being performed in accordance with the security check code corresponding to the information obtained based on the identification information.

7. An information recording medium manufacturing apparatus comprising:

content file obtaining means for obtaining content files storing content to be recorded on an information recording medium;

content code file obtaining means for obtaining content code files storing a plurality of content codes comprising security check codes and a security check processing program that executes the security check codes when the content is used; and

recording means for recording the content files obtained by the content file obtaining means and the content code files obtained by the content code file obtaining means on the information recording medium,

wherein the content code file obtaining means obtains a plurality of content code files comprising:

at least one first content code file that stores first content codes from the plurality of content codes that do not depend on types of information processing apparatuses or content usage applications and do not depend on the content,

at least one second content code file that stores second content codes from the plurality of content codes that depend on the types of information processing apparatuses or content usage applications, and

at least one third content file that stores third content codes from the plurality of content codes for determining the types of information processing apparatuses or content usage applications, and wherein the type of the information processing apparatuses comprises manufacturers of the information processing apparatuses.

8. The information recording medium manufacturing apparatus according to claim 7, wherein the content code file obtaining means obtains a code file storing a code for determining the type of an information processing apparatus or content usage application.

9. An information recording medium comprising:

content files storing content; and

content code files storing a plurality of content codes each including a security check processing program to be executed when the content is used by an information processing apparatus,

wherein the content code files comprise a plurality of content code files comprising:

at least one first content code file that stores first content codes from the plurality of content codes that do not depend on a type of the information processing apparatus or a content usage application and do not depend on the content,

at least one second content code file that stores second content codes from the plurality of content codes that depend on the type of the information processing apparatus and port information on the information processing apparatus, and

at least one third content file that stores third content codes from the plurality of content codes for determining the type of the information processing apparatus or the content usage application.

10. The information recording medium according to claim 9, wherein the content code files include a code file storing a code for determining a type of an information processing apparatus or a content usage application.

11. An information processing method for use in an information processing apparatus and for performing data processing using data comprising content and content codes recorded on an information recording medium, the method comprising:

with at least one processor, performing the data processing by:

in response to a determination that the content codes recorded on the information recording medium include a request to obtain from memory of the information processing apparatus identification information related to a player certificate or a port status of the information processing apparatus, obtaining the identification information;

selecting, from security check codes included in the content codes, a security check code corresponding to the identification information comprising certificate information of the information processing apparatus and configuration information that comprises a port status of the information processing apparatus; and

when using the content recorded on the information recording medium, conducting a security check for the information processing apparatus in accordance with the selected security check code.

12. The information processing method according to claim 11, wherein, in the execution of the data processing,

the player certificate stored in a storage unit of the information processing apparatus is obtained and an integrity of the player certificate is verified, and

when the integrity of the player certificate is verified, the identification information is obtained from information recorded on the player certificate.

13. The information processing method according to claim 11, wherein, in the execution of the data processing, at least one of port attribute information, memory area size information, and data processing mode information concerning the information processing apparatus is obtained as the configuration information, and security check corresponding to the obtained configuration information is conducted.

14. The information processing method according to claim 11, wherein, in the execution of the data processing, device information managed by an operating system is obtained

when obtaining the configuration information, and security check corresponding to the obtained device information is conducted.

15. The information processing method according to claim **11**, wherein, in the execution of the data processing, when the configuration information is updated, the updated configuration information is obtained, and security check corresponding to the updated configuration information is conducted.

16. The information processing method according to claim **11**, wherein, in the execution of the data processing, data processing which accompanies content data replacement processing using a conversion table included in the content codes is performed, the data processing being performed in accordance with the security check code corresponding to the information obtained based on the identification information.

17. An information recording medium manufacturing method for use in an information recording medium manufacturing apparatus, comprising:

obtaining content files storing content to be recorded on an information recording medium;

obtaining content code files storing a plurality of content codes comprising a security check processing program to be executed when the content is used by an information processing apparatus or a content usage application, the content code files comprising:

at least one first content code file that stores first content codes from the plurality of content codes that do not depend on a type of the information processing apparatus or the content usage application and do not depend on the content,

at least one second content code file that stores second content codes from the plurality of content codes that depend on the type and a port status of the information processing apparatus or the content usage application, and

at least one third content file that stores third content codes from the plurality of content codes for determining the type of the information processing apparatus or the content usage application; and

recording the obtained content files and the obtained content code files on the information recording medium.

18. The information recording medium manufacturing method according to claim **17**, wherein, in the execution of obtaining the content code files, a code file storing a code for determining the type of the information processing apparatus or the content usage application is obtained.

19. An information recording medium manufacturing apparatus comprising:

a content file obtaining unit configured to obtain content files storing content to be recorded on an information recording medium;

a content code file obtaining unit configured to obtain content code files storing a plurality of content codes comprising a security check processing program to be executed when the content is used by an information processing apparatus or a content usage application, the content code files comprising:

at least one first content code file that stores first content codes from the plurality of content codes that do not depend on a type of the information processing apparatus or the content usage application and do not depend on the content,

at least one second content code file that stores second content codes from the plurality of content codes that depend on the type of the information processing apparatus or the content usage application, and

at least one third content file that stores third content codes from the plurality of content codes for determining the type of the information processing apparatus or the content usage application; and

a recorder configured to record the content files obtained by the content file obtaining unit and the content code files obtained by the content code file obtaining unit on the information recording medium.

20. The information processing apparatus according to claim **1**, wherein the data processor, in response to a determination that the content codes do not include the request, causes playback of the content.

21. The information processing apparatus according to claim **1**, wherein the data processor performs the determination that the content codes include the request, the determination comprising obtaining from at least one content code file storing at least one content code from the content codes a code for determining the identification information.

22. The information processing method according to claim **11**, further comprising, in response to a determination that the content codes do not include the request, causing playback of the content.

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